

USER MANUAL



Spirit® 1040-16-HE

Spirit® 1040-16-HE-SHG

Spirit® 1040-30-HE

Spirit® 1040-30-HE-SHG

Content

Content	2
Welcome	5
1 Introduction	6
2 Warning Convention	7
3 Safety	8
3.1 General Safety Instructions	8
3.2 Laser Safety Instructions	9
3.3 Types of Labels	11
3.3.1 Laser Safety Labels	11
3.3.2 Electrical Type Labels	13
3.4 Safety Devices	14
3.4.1 Emission Indicators	14
3.4.2 Key switch	14
3.4.3 Mechanical Laser Shutter	14
3.4.4 Laser Safety Shutter	15
3.4.5 Laser Safety Interlock	15
3.5 Electromagnetic Compatibility (EMC)	15
3.6 General Warnings and Recycling	18
4 General Conditions for Laser System	19
4.1 Transport Conditions	19
4.2 Storage Conditions	19
4.3 Laser Operating Conditions	19
5 Equipment Installation	21
5.1 Unpacking the Laser System	21
5.1.1 Inspection of Shipping Boxes	21
5.1.2 Unpacking the Laser System	21
5.1.3 Warranty Seals	21
5.2 Installation of the Laser System	22
5.2.1 Setting up the Laser Head	22
5.2.2 Setting up the Electronics Rack	22
5.2.3 Chiller	23
5.2.4 Installation of the Cable System	25
5.2.5 Installation Checklist	28
6 Communication	29
6.1 Connecting the Controller to a CANopen Network	29
6.1.1 CANopen protocols	29

6.1.2	CAN In, CAN Out.....	29
6.1.3	Node ID.....	29
6.1.4	Baud rate	30
6.2	Using the CANopen interface	30
6.2.1	Network Management – Application State Machine.....	30
6.2.2	User / Service / Factory Profiles.....	30
6.2.3	Control, PDO Mapping	30
6.2.4	SDO Parameter (Service Data Objects)	30
6.2.5	Error Handling via EMC Message	31
6.3	Using the TCP/IP interface.....	31
6.3.1	Overview	31
6.3.2	Command	31
6.3.3	Interface Settings	32
7	Operation	34
7.1	Operating the laser system.....	34
7.2	Graphical User Interface - HMT	35
7.2.1	“System OFF”	35
7.2.2	“Standby”	36
7.2.3	“System ON”	37
7.2.4	“Advanced View”	38
7.2.5	Shutter Control and Laser Safety	38
7.2.6	Repetition Rate Adjustment.....	39
7.2.7	Power Control	40
7.2.8	Laser Power Modulation using “Analog In”	41
7.2.9	Pulse Enable and SYNC Out	43
7.2.10	Variable Pulse Duration (optional).....	44
7.2.11	Burst Mode (optional)	45
7.2.12	Power Lock	46
7.2.13	Auto Tune.....	46
8	Maintenance and Service.....	47
8.1	Maintenance.....	47
8.1.1	Changing the Active Laser Purge System (ALPS)	48
8.2	Service Contact	51
9	Diagnostics and Troubleshooting	52
9.1	Error Codes	52
9.2	Troubleshooting.....	55
10	Specifications	59
10.1	System Drawings Laser Head.....	59

10.2	System Drawings Electronic Rack	60
10.3	Optical, Mechanical and Electrical Specifications 16 W HE-SHG	61
10.4	Optical, Mechanical and Electrical Specifications 30 W HE-SHG	62
11	Appendix A1: Spirit CANopen Communication Overview.....	63
12	Appendix A2: Spirit TCP/IP Communication Overview.....	68
13	APPENDIX A3: User Control Interface	74
14	APPENDIX A4: China RoHS.....	75
15	Notes	76

Welcome

Thank you for your purchase of Spectra-Physics instruments!

The Spectra-Physics / High Q Laser GmbH Spirit® HE femtosecond laser is a solid-state, high repetition rate one box ultrafast amplifier. This manual contains information for the safe installation, operation, and maintenance of your laser system.

For further information concerning the Spirit® laser system, please contact a local service representative.



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Article number for **USER MANUAL “Spirit® 1040-16-HE, Spirit® 1040-16-HE-SHG, Spirit® 1040-30-HE, Spirit® 1040-30-HE-SHG”**:

AN11004 (Rev. 12)

1 Introduction



The Spectra-Physics Spirit® systems bring more capability, greater process range, and better value to customers looking for uncompromising performance and reliability in lasers for micro-machining and other applications. The laser's high pulse energy (up to 120 μJ) and high average power ($>30\text{ W}$) enable femtosecond micromachining applications with high throughput and the best quality. Pulse energy and repetition rate adjustability (single shot to 1 MHz) make the Spirit HE an ideal laser source for medical device manufacturing, machining of transparent and brittle materials and high aspect ratio drilling. The optional, high efficiency Second Harmonic Generation (SHG) module is integrated in the laser head and enables the fabrication of smaller structures which require higher resolution and therefore a smaller spot size. The end user can easily switch between the 1040 nm and 520 nm outputs via the user interface.

APPLICATIONS

The Spirit® systems were designed specifically to enable applications demanding precision laser micro-machining to go faster, with higher quality, and at lower cost, with less downtime and lower maintenance costs. These applications include, among others:

- Femtosecond micromachining
- Material processing
- Medical device fabrication
- Cutting of transparent brittle materials
- Microsurgery
- Time resolved femtosecond spectroscopy
- Optical parametric amplifier pumping

2 Warning Convention

The following symbols are used throughout this manual to draw the user's attention to situations or procedures that require extra attention. They warn of health hazards, damage to equipment, sensitive procedures, and exceptional circumstances. All messages are set apart by a thin line above and below the text as shown here.

WARNING SYMBOLS

Danger: Indicates an imminently hazardous situation which will result in death or serious danger, if not avoided.	
Warning: Indicates a potentially hazardous situation or an unintended use which could result in death or serious danger, if not avoided.	
Caution: Indicates a potentially hazardous situation or an unintended use which may result in minor or moderate injury and/or appreciable material, financial and environmental damage, if not avoided.	
Laser Radiation: Indicates hazardous situations regarding Laser Class IV Products which will result in serious injuries, if not avoided.	
Important: Paragraphs which must be adhered to in practice as they enable the product to be used in a technically correct and efficient manner.	
Information: This paragraph gives additional information or highlights a point of special interest about the topic under discussion.	

3 Safety

This chapter provides important information on general safety and laser safety and Spectra-Physics recommends that the user read it carefully.

3.1 General Safety Instructions

Person responsible for the product

The person responsible for this product has the following duties:

- To understand the safety instructions on the product and the instructions in the user manual.
- To ensure that it is used in accordance with the instructions.
- To ensure that all users understand these directions and adhere to them.
- To be familiar with local regulations relating to safety and accident prevention.
- To inform Spectra-Physics immediately, if the product becomes unsafe.
- To ensure that the national laws, regulations and conditions for the operation of lasers are respected.

Intended Use

- Operation according to the descriptions in this user manual.
- Operation under the environmental conditions defined in this user manual.

Reasonably foreseeable misuse

- Use of the product without instruction.
- Use outside of the intended use and limits.
- Disabling safety systems.
- Removal of hazard notices.
- Opening the product using tools, for example a screwdriver, unless permitted for certain functions.
- Modification or conversion of the product.
- Use after misappropriation.
- Use of the product with recognizable damages or defects.
- Use with accessories from other manufacturers without prior explicit approval of Spectra-Physics.

Hazards of Use

The user must be aware of the following risks and dangers that may occur during incorrect operation:

Caution:

Watch out for erroneous operation, if the product has been dropped or has been misused, modified, stored for long periods or transported.

Precautions:

Periodically carry out tests to assess the performance of the product indicated in the user manual.



Warning:

Do not open the laser system! Only Spectra-Physics authorized persons are entitled to open this product.

If you open the product, either of the following actions may cause you to receive an electric shock:

- Touching live components

Using the product after incorrect attempts were made to carry out repairs.



Warning:

Lift and/or carry the product with a minimum of two persons.

Because of the weight, lifting and/or carrying the laser system and/or its components (e.g. the electronics rack) may lead to injury.



3.2 Laser Safety Instructions

The Spirit® laser system is a Class IV laser that emits laser radiation that can permanently damage eyes and skin. This section contains information about laser safety according to international standard IEC 60825-1: 2014. It contains information about these hazards and offers suggestions on how to safeguard against them.



National laws and local regulations could impose more stringent instructions for the safe use of lasers than IEC 60825-1: 2014.

The Spectra-Physics Spirit® laser is a Class IV—High-Power Laser whose beam is, by definition, a safety and fire hazard. Take precautions to prevent accidental exposure to both direct and reflected beams. Diffused as well as specular beam reflections can cause severe eye or skin damage. The laser emerging from the laser aperture is *invisible and/or visible*.

Safety Precautions for Class IV High-Power Lasers

- Wear protective eyewear at all times. Eyewear selection depends on the wavelength and intensity of the radiation, the conditions of use, and the visual function required.
- Maintain a high ambient light level in the laser operation area so that the eye's pupil remains constricted, reducing the possibility of retinal damage.
- Avoid looking at the output beam; even diffused reflections are hazardous.
- Avoid blocking the output beam or its reflections with any part of the body.
- Establish a controlled access area for laser operation. Limit access to personnel trained in the principles of laser safety.
- Enclose beam paths wherever possible.
- Post prominent warning signs near the laser operating area.

- Install the laser so that the beam is below eye level.
- Set up shields to prevent any unnecessary specular reflections or beams from escaping the laser operation area.
- When the target is not present, set up a beam dump to capture the laser beam and prevent accidental exposure.
- Do not block off access to any disconnect device such as the key-switch, shutter control, power cord, etc.

Warning:

Operating a class 4 laser without sufficient instruction or training can put employees or others at risk of severe injury.

Precautions:

Sufficient instruction or training should be given in order that employees have the necessary understanding to avoid placing themselves and others at unacceptable risk. Such instruction and training should be commensurate with the type of hazard and appropriate for the employees concerned.

Only qualified and trained persons shall be assigned to install, adjust and operate the product.



Warning:

It is forbidden to change any configuration in any part of the whole system (electronics rack, power and control cables, water tubes and laser head). For the reason that direct or scattered laser light can get out of the laser housing in an uncontrolled way, it is also forbidden to open any lid of the laser head.

The warranty seal on the main laser lid must not be broken or damaged.



Important:

The laser head is equipped with a white **laser warning LED**. This LED is on when radiation is generated by the laser source even if the safety shutter is closed.



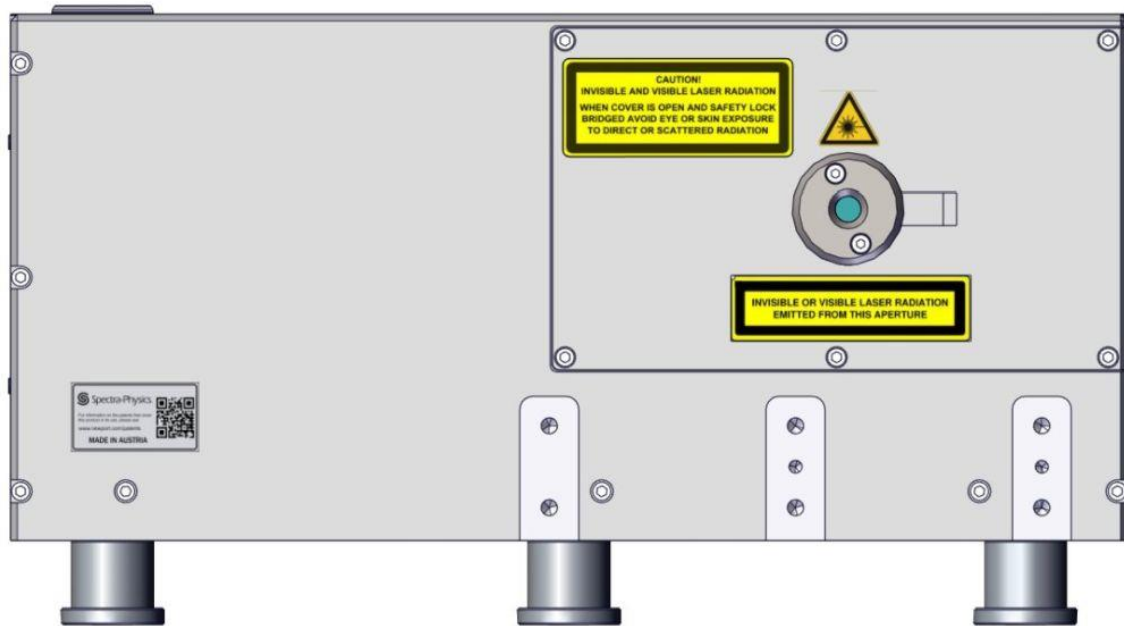
3.3 Types of Labels

3.3.1 Laser Safety Labels

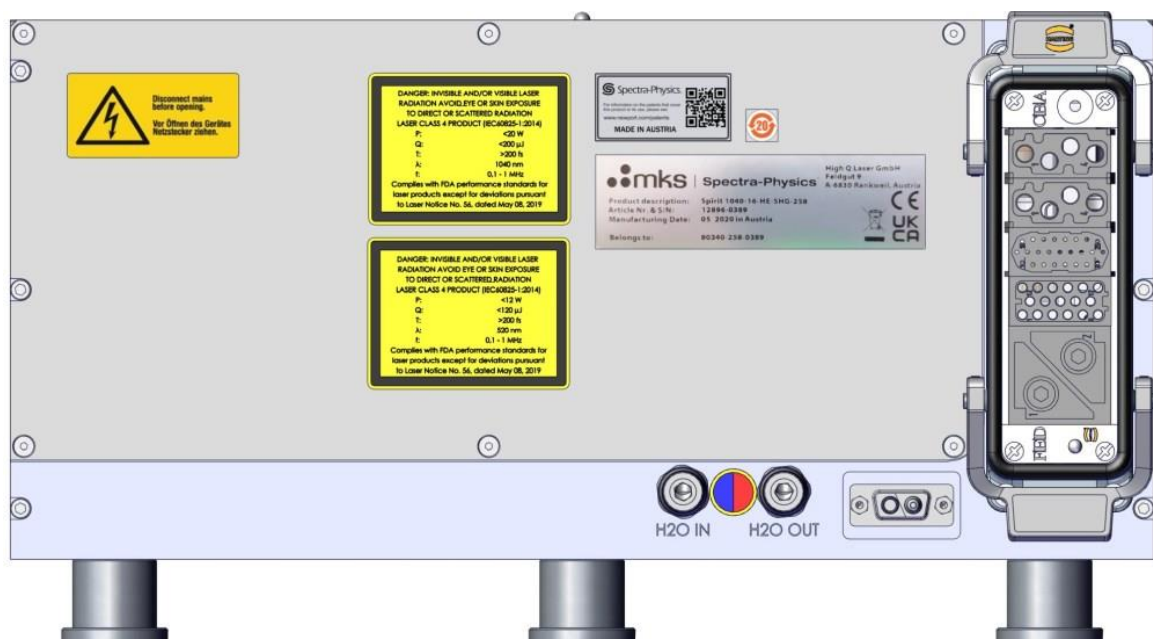
The laser head is equipped with laser safety labels which provide information about the maximum ratings for each wavelength of the laser.

Laser head label placement

Laser head – front view



Laser head – rear view



Laser head labels

Visible or invisible laser radiation may be emitted from the laser aperture.

**INVISIBLE OR VISIBLE LASER RADIATION
EMITTED FROM THIS APERTURE**

This laser safety label provides information about the maximum ratings for the laser for invisible light.

SPIRIT 1040-16 HE

DANGER: INVISIBLE AND/OR VISIBLE LASER
RADIATION AVOID EYE OR SKIN EXPOSURE
TO DIRECT OR SCATTERED RADIATION
LASER CLASS 4 PRODUCT (IEC60825-1:2014)

P: <20 W
Q: <200 µJ
T: >200 fs
λ: 1040 nm
f: 0,1 - 1 MHz

Complies with FDA performance standards for
laser products except for deviations pursuant
to Laser Notice No. 56, dated May 08, 2019

SPIRIT 1040-30 HE

DANGER: INVISIBLE AND VISIBLE LASER RADIATION
WHEN OPEN AVOID EYE OR SKIN EXPOSURE
TO DIRECT OR SCATTERED RADIATION
LASER CLASS 4 PRODUCT (IEC60825-1:2014)

P: <35 W
Q: <200 µJ
T: >200 fs
λ: 1040 nm
f: 0,2 - 1 MHz

Complies with FDA performance standards for
laser products except for deviations pursuant
to Laser Notice No. 56, dated May 08, 2019

This laser safety label provides information about the maximum ratings for the laser for visible light.

SPIRIT 1040-16 HE-SHG

DANGER: INVISIBLE AND/OR VISIBLE LASER
RADIATION AVOID EYE OR SKIN EXPOSURE
TO DIRECT OR SCATTERED RADIATION
LASER CLASS 4 PRODUCT (IEC60825-1:2014)

P: <12 W
Q: <120 µJ
T: >200 fs
λ: 520 nm
f: 0,1 - 1 MHz

Complies with FDA performance standards for
laser products except for deviations pursuant
to Laser Notice No. 56, dated May 08, 2019

SPIRIT 1040-30 HE-SHG

DANGER: INVISIBLE AND VISIBLE LASER RADIATION
WHEN OPEN AVOID EYE OR SKIN EXPOSURE
TO DIRECT OR SCATTERED RADIATION
LASER CLASS 4 PRODUCT (IEC60825-1:2014)

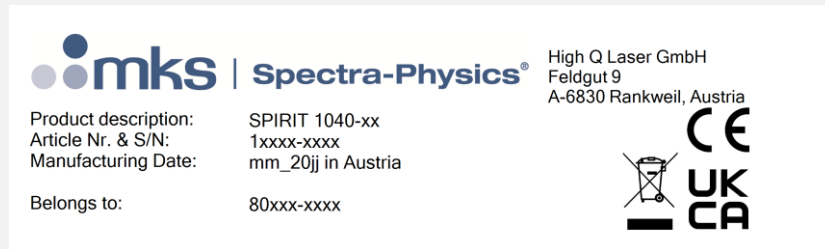
P: <20 W
Q: <120 µJ
T: >200 fs
λ: 520 nm
f: 0,2 - 1 MHz

Complies with FDA performance standards for
laser products except for deviations pursuant
to Laser Notice No. 56, dated May 08, 2019

The laser safety label shown indicates where the laser beam exits the laser and laser light is emitted. Each single exit port is indicated with one safety label.



3.3.2 Electrical Type Labels



This type label on the back cover of the laser head includes the following information:

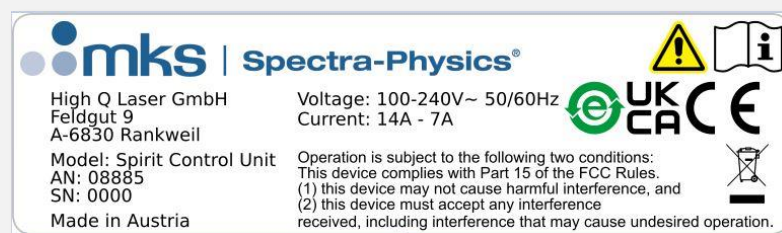
- Address of mks/Spectra Physics
- Article and serial number of laser head
- Manufacturing country and date
- Overall system and serial number

The Article No. 80340 represent the laser system SPIRIT 1040-16 HE. With the following options the Article No. varies to 8034x (x =1 to 7): Second Harmonic Generation (-SHG), Clear Shape (-CS) and Variable Pulse Duration (-fs2ps). The same applies for the SPIRIT 1040-30 with the Article No. 80350 and 8035x for selected options.

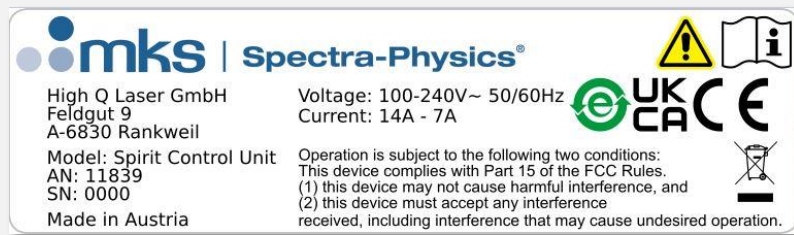
The label on the back of the electronics rack includes the following information:

- mks/Spectra-Physics company logo
- Company address
- Model description of electronics rack
- Supply voltage and current rating
- Serial number of electronics rack
- CE mark

The following label present an example of the label on the stand-alone electronic rack AN08885 delivered with the Spirit 16 W HE laser system.

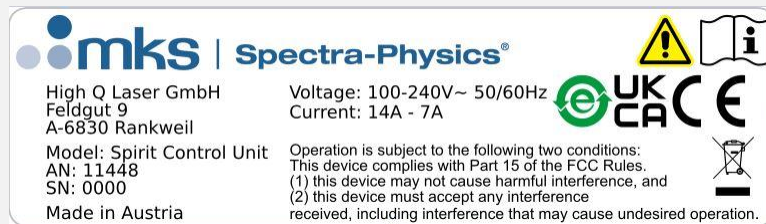


While the following one represent an example of the electronic rack label present on the stand-alone electronic rack AN11839 delivered with the Spirit 30 W laser system.

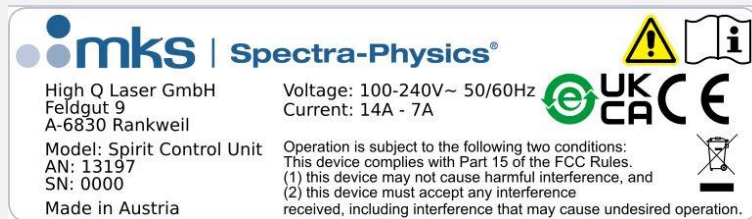


Optional SPLIT Rack

The following label present an example of the label on the SPLIT electronic rack AN11448 delivered with the Spirit 16 W HE laser system (optional).



While the following one represent an example of the electronic rack label present on the SPLIT electronic rack AN13197 delivered with the Spirit 30 W laser system (optional).



3.4 Safety Devices

3.4.1 Emission Indicators

The laser head as well as the laser controller have emission indicators. A white light emitting diode (LED) on top of the laser head and two redundant indicators on the front panel of the laser controller. They turn on prior to actual potential laser emission.

3.4.2 Key switch

The key switch on the front panel of the electronic rack prevents unauthorized use of the laser when it is turned to the 'disabled' position and the key is removed. Placing this switch in the 'enabled' position allows the laser to be turned on.

3.4.3 Mechanical Laser Shutter

A mechanical shutter is mounted at the laser output aperture. No laser output is available when the mechanical shutter is closed. The shutter is closed at delivery of the laser system and should remain closed during the initial installation and setup procedure. They can only be opened manually.

3.4.4 Laser Safety Shutter

The laser system has an integrated redundant safety shutter relay (type: PSR-SCP- 24DC/ESD/4X1/30 Phoenix Contact). The redundant safety shutter relay has one input (accessible via pins 9 and 17 of the user control interface), two not time delayed outputs (not accessible to the user) and two-time delayed outputs (accessible via pins 3/4 and 6/7 of the user control interface) with a preset delay time of 1 s. In order to open the electro-mechanical shutter, the pins 9 and 17 of the user control interface must either be wired to a normally closed switch or be short circuited. In that case, the shutter can be opened via software command. The time delayed contacts are available for additional safety monitoring at the back of the electronic rack - User Control Interface (pins 3/4 and 6/7).

Please refer also to APPENDIX A3: User Control Interface.

3.4.5 Laser Safety Interlock

Interlock pins 1 and 10 of the laser control interface on the back of the electronic rack must be connected for normal system operation (see APPENDIX A3: User Control Interface). These pins can be used in an interlock circuit to terminate laser emission when a normally closed safety switch wired to these pins is opened. The safety switch can be attached to an access point, such as an entry panel, that might be opened unexpectedly.

Important:

For laser operation, the two contacts (interlock pins 1 and 10) of the user control interface (APPENDIX A3: User Control Interface) **MUST** either be wired to a normally closed safety switch or be short circuit. In order for the shutter to be opened, the two contacts (pins 9 and 17) of the user control interface (APPENDIX A3: User Control Interface) **MUST** either be wired to a normally closed switch or be short circuit.



3.5 Electromagnetic Compatibility (EMC)

Description

The term Electromagnetic Compatibility is taken to mean the capability of the product to function smoothly in an environment where electromagnetic radiation and electrostatic discharges are present, and without causing electromagnetic disturbances to other equipment.

Warning:

Electromagnetic radiation can cause disturbances in other equipment.



Although the product meets the strict regulations and standards which are in force in this respect, Spectra-Physics / High Q Laser GmbH cannot completely exclude the possibility that other equipment may be disturbed.

Class A devices

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules.

These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment.

This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications.

Operating this equipment in a residential area is likely to cause harmful interference.

Warning:

This laser system is a Class A EMC product. This product can cause disturbances in domestic environment. In case of disturbances the user has to take adequate measures.

**Caution:**

There is a risk that disturbances may be caused in other equipment if the product is used in conjunction with accessories from other manufacturers (e.g. personal computers, power supplies).

Precautions:

Use only the equipment and accessories recommended by Spectra-Physics. When combined with the product, they meet the strict requirements stipulated by the guidelines and standards. When using accessories from other manufacturers, pay attention to the information about electromagnetic compatibility provided by the manufacturer.

**Caution:**

Disturbances caused by electromagnetic radiation can result in faulty functioning. Although the product meets the strict regulations and standards which are in force in this respect, Spectra-Physics cannot completely exclude the possibility that the product may be disturbed by very intense electromagnetic radiation (e.g. radio transmitters).

Precautions:

Check the plausibility of the product performance obtained under these conditions.

**Warning:**

Changes or modifications not expressly approved by Spectra-Physics for compliance could void the user's authority to operate the equipment.



CE declaration of conformity

Hereby, Spectra-Physics declares that the product/s is/are in compliance with the essential requirements and other relevant provisions of the applicable European Directives. For obtaining the declaration of conformity please contact Spectra-Physics.

We,

Spectra-Physics / High Q Laser GmbH

Feldgut 9
6830 Rankweil
Austria

declare under sole responsibility that the

Spirit® HE Family

following the provisions of Directive(s)

2014/30/EU EC Electromagnetic compatibility (EMC)
2014/35/EU Low voltage (LVD)
2011/65/EU Restriction of hazardous substances (RoHS)

to which this declaration relates, is in conformity with the following standards:

EN 60825-1:2014
EN 61010-1:2010+A1:2016
EN 61326-1:2012



Information:

The laser system follows the below mentioned guidelines:

- Overvoltage category II (Überspannungskategorie II)
- Voltage fluctuation: $\pm 10\%$
- Maximum altitude: 3 000 m



3.6 General Warnings and Recycling

Warning Sign “High Voltage”:

This symbol is intended to alert the operator to the presence of dangerous voltages associated with the product that may be of sufficient magnitude to constitute a risk of electrical shock.

Information:

Due to high voltage capacitors dangerous voltages can persist after disconnecting the main power.



Warning Sign “Grounding”:

If the product is not connected to ground, death or serious injury can occur!

Precautions:

To avoid electric shock power cable and power outlet must be grounded. The product has to be connected only to a supply network with a ground wire.



Important:

The product must not be disposed with household waste!

It must be disposed appropriately in accordance with the national regulations in force in your country.

For additional questions regarding disposal of the laser system please do not hesitate to get in contact with the company Spectra-Physics.



4 General Conditions for Laser System

Important:

This section provides detailed information about the required conditions for transportation, storage and operation of the system.

This laser system is designated to be used in indoor laboratories for trained users only!

For unpacking and installing the laser system, you will be assisted by a factory expert or service engineer from Spectra-Physics.



4.1 Transport Conditions

The packaging case is designed to prevent damages to the laser system. This box is equipped with shock indicators which have to be checked at the arrival of the system. If the indicators are activated or any damage to the shipping box can be observed, the freight forwarded, and Spectra-Physics have to be informed immediately.

The transport conditions are according to EN-ISO 15004-1:2009-7:

- Temperature: 0°C – 40°C
- Rel. Humidity: < 65%, non-condensing, non-freezing
- Air pressure: 500 – 1060hPa (non-condensing)

Important:

To ensure non-condensing conditions, a desiccant was/must be added in the shipping box.



4.2 Storage Conditions

For storage of the laser system the following conditions according to EN-ISO 15004-1:2009-7 must be ensured:

- Temperature: 0°C – 40°C
- Rel. Humidity: < 65%, non-condensing, non-freezing
- Air pressure: 500 – 1060hPa (non-condensing)

4.3 Laser Operating Conditions

The operating conditions for the system are specified according to EN-ISO 15004-1:2009-7 as:

- Temperature: 18°C – 30°C
- Temp. gradient: < 5°C/h
- Rel. Humidity: < 65%, non-condensing
- Air pressure: 700 – 1060hPa (non-condensing)

Information:

It is recommended to mount the laser head on three pedestals using three clamping forks.



5 Equipment Installation

This section provides detailed instructions for installing the Spirit® laser system.

5.1 Unpacking the Laser System

5.1.1 Inspection of Shipping Boxes

Before unpacking the laser system please make sure that the shipping boxes do not show any damage and that shock indicators have not been activated during transportation.

Any damage must be reported immediately to the freight forwarder and to Spectra-Physics.

5.1.2 Unpacking the Laser System

For opening the shipping box please remove the cover. The side walls can also be removed from the base plate to get better access to the electronics rack and the laser head.

Caution:

The laser head and especially the electronics rack are heavy. For this reason, both the laser head and electronics rack must be carried by a minimum of two.



5.1.3 Warranty Seals

After mounting the laser head, please verify that the warranty seal on the main laser lid is not broken. A broken or damaged warranty seal may indicate an unauthorized opening of the laser case and causes an expiration of the warranty time.

The warranty seal was affixed to the laser by Spectra-Physics and ensures that a quality inspection has been performed.



Important:

Opening the laser system or breaking the warranty seal will void the warranty!
In case of a broken warranty seal Spectra-Physics has to be informed prior to initial system operation!



5.2 Installation of the Laser System

5.2.1 Setting up the Laser Head

The Spirit® laser is designed for operation indoors. The laser has an operating temperature range of 18 to 30°C. Protection from process debris and airborne particulates is recommended.

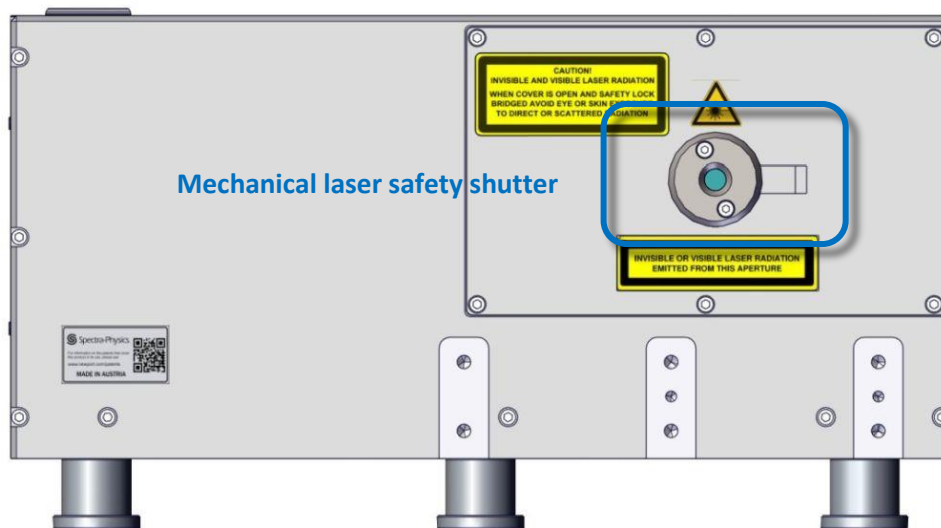
The bottom of the laser head's baseplate has three pedestals designed for mounting the laser head on a suitable flat surface. Use three of the provided clamps, either imperial (1/4-20 bolts) or metric (M6 bolt), to fix the pedestals on the flat surface.

Information:

To allow laser emission it is necessary to ensure that the mechanical shutter is opened. The mechanical shutter is located at the front panel of the laser can only be opened manually.



Mechanical laser safety shutter at the front of the laser head



5.2.2 Setting up the Electronics Rack

A stand-alone electronics rack is part of the laser system (see chapter 5.2.2.1). Optionally, if you have chosen a Split-Rack configuration, see chapter 5.2.2.2.

In either case, please ensure that the electronics rack is placed so that any accident, for example due to tripping over cables, is avoided. This may cause damage to the laser system and/or harm to personnel.

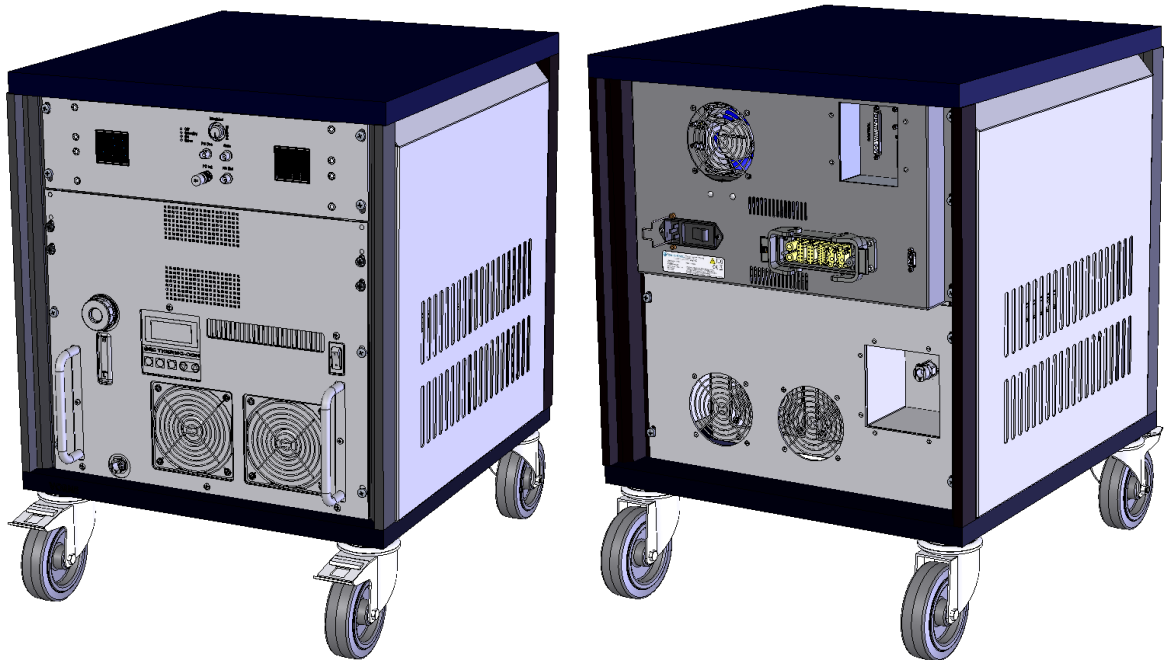
Caution:

The electronics rack must be placed on solid ground and the back of the rack must be placed at least 50 cm away from any wall or other object to ensure sufficient air circulation for the water cooling unit (chiller).
Make sure that the utility main power switch (I / 0) is accessible to allow the unit to be powered on and off.



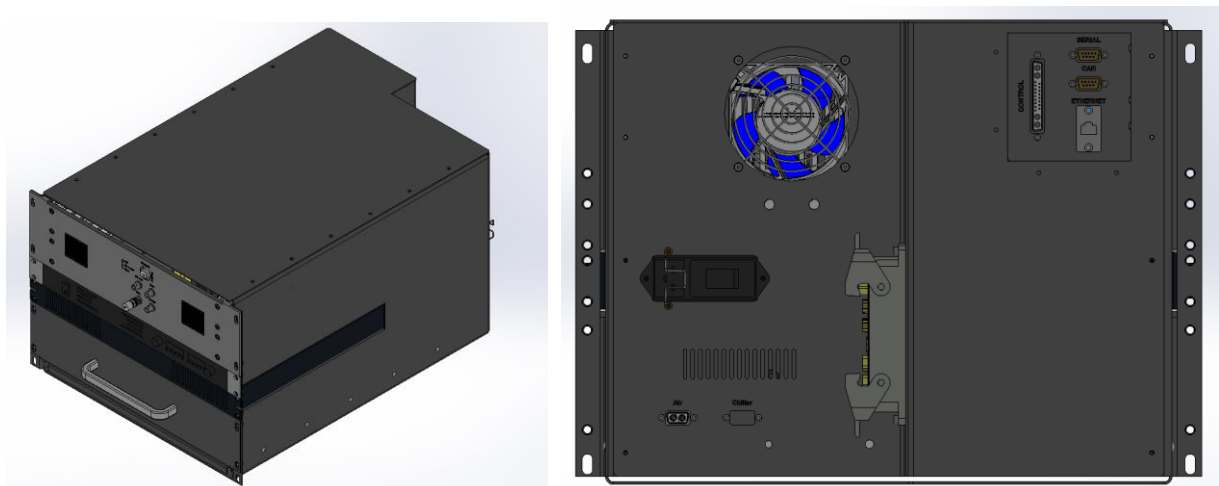
5.2.2.1 Stand-Alone Electronics Rack

The stand-alone electronics rack contains the chiller as well as the laser control unit. The electronics rack must be placed near the laser head due to the limited length of the electronics control cable.



5.2.2.2 SPLIT-Rack (OPTIONAL)

The Split-Rack is an 8 U rack height and is 19" rack mountable. When mounting the Split-Rack it must be ensured that air flow for cooling is not blocked and that there is sufficient clearance for cabling and tubing. The main power switch should be accessible.



5.2.3 Chiller

Spectra-Physics provides a chiller that is suitable for use with the laser system. If you ordered the chiller from the factory with the laser, it meets these requirements. For specific information about the chiller shipped with the system, please consult the included chiller's user manual.

Important:

Antifrogen® L by Clariant and distilled water are the only approved coolants for use with the Spirit® laser. Do NOT use tap or deionized (DI) water.

Reference address for Clariant's Antifrogen® L: Please visit www.clariant.com for further information on how to order in your country.



Important:

Do not switch off the chiller unit during operation of the laser system (main switch at the front panel of the chiller) as this may result in damage to the laser system.



5.2.3.1 Closed-loop water chiller: SMC (model: HECR008)



The laser head's temperature is controlled and stabilized by a closed-loop water chiller by **SMC** (model: **HECR008**).

Filling Procedure

- 1) Consult the chiller's user manual and follow the manufacturer's suggested operation requirements.
- 2) Remove the tank cap of the chiller.
- 3) Put the provided funnel into the tank opening for an easier filling.
- 4) Pour distilled water or Antifrogen® L in addition with distilled water into the chiller reservoir (300ml or 600ml).

Necessary amount of Antifrogen® L:

- Water hose length = **3m or 5m** → **300ml** of Antifrogen L
- Water hose length = **10m** → **600ml** of Antifrogen L

- 5) Close the tank cap of the chiller.
- 6) Connect the coolant hoses between the laser head and chiller.

The direction of the coolant flow is important for proper operation of the laser system. Plug the red and blue hoses (3m or 10m) into the outlets and inlets of the chiller and the laser head according to

the color markings. For more detailed information see also chapter **Fehler! Verweisquelle konnte nicht gefunden werden..**

- 7) Turn the chiller on. As the fluid in the reservoir drops, continue to fill it with distilled water until it no longer drops.
- 8) Verify that there are no water leaks from the chiller, the hoses, or the laser. **Make sure the reservoir does not empty!**
- 9) Verify that the chiller is set to 25°C. If the chiller was ordered from Spectra-Physics, it was pre-set at the factory to operate at 25°C.

5.2.3.2 Water-water Chiller (OPTIONAL)

5.2.3.2.1 SMC (model: HECR008-W5)

The laser head's temperature can be optionally controlled and stabilized with a water-water chiller.

For installing the chiller system, please consult the chiller's user manual and follow the manufacturer's suggested operation requirements.

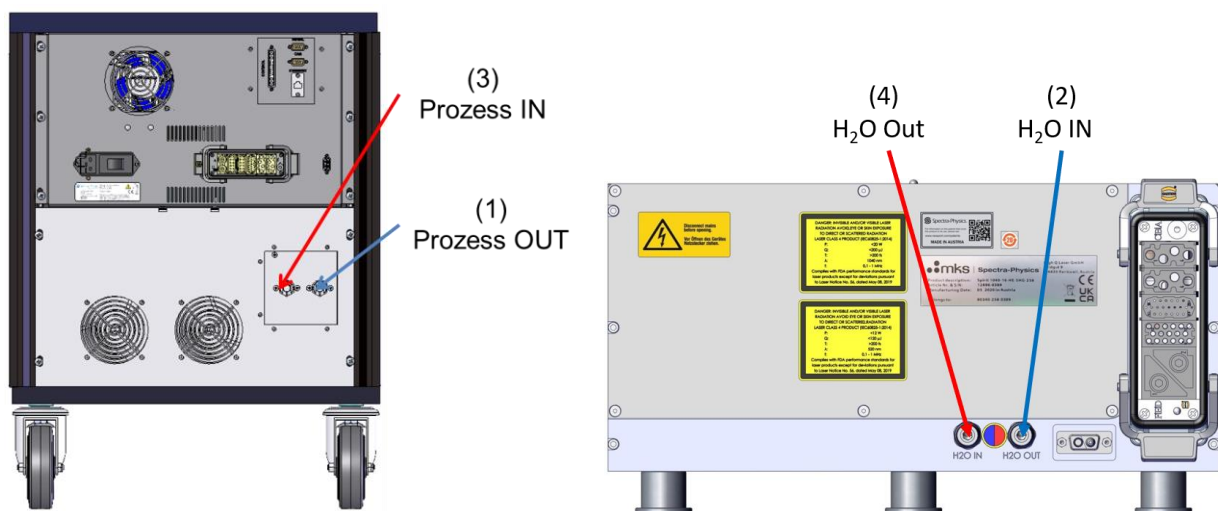


5.2.4 Installation of the Cable System

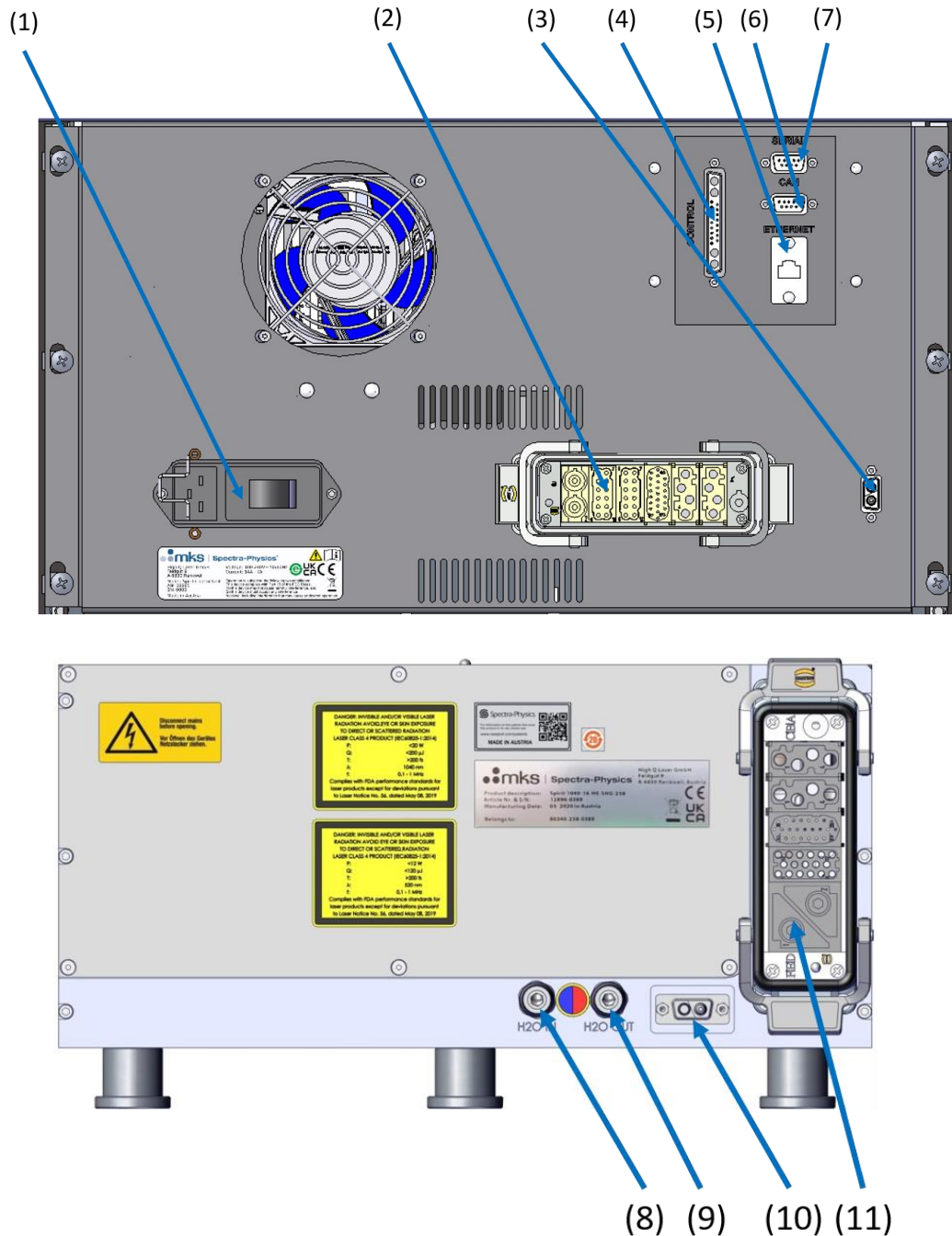
This section describes how to connect the various components of the laser system.

5.2.4.1 Stand Alone Electronics Rack

For the water cooling circuit, the water connectors of the electronics rack and the laser head must be connected with the two provided water tubes. The first tube (**blue water tube**) connects the **Process Out** of the electronics rack (1) with the **H₂O In** of the laser head (2). The second water tube (**red water tube**) including the attached water filter box connects the **Process In** of the electronics rack (3) with the **H₂O Out** of the laser head (4).



Before setting up cable connections between electronics rack and the laser head, ensure proper grounding of the electronics rack through the main plug **(1)**. Do NOT switch on the electronic rack using the main switch (I / O) until all cables and water hoses are connected.



- 1) Attach the power cable to the electronic rack **(1)** and plug the line cord into a grounded outlet.
- 2) Attach the control cable between the electronic rack **(2)** and the laser head **(11)**.
- 3) If the delivered Spirit® laser system is equipped with an optional SHG compartment, the integrated SHG box has to be flushed with purified air. Therefore, attach the air purifying cable between electronic rack **(3)** and laser head **(10)**.
- 4) The user control interface has to be attached to the electronic rack **(4)**. The connector plug configuration is described in APPENDIX A3: User Control Interface.

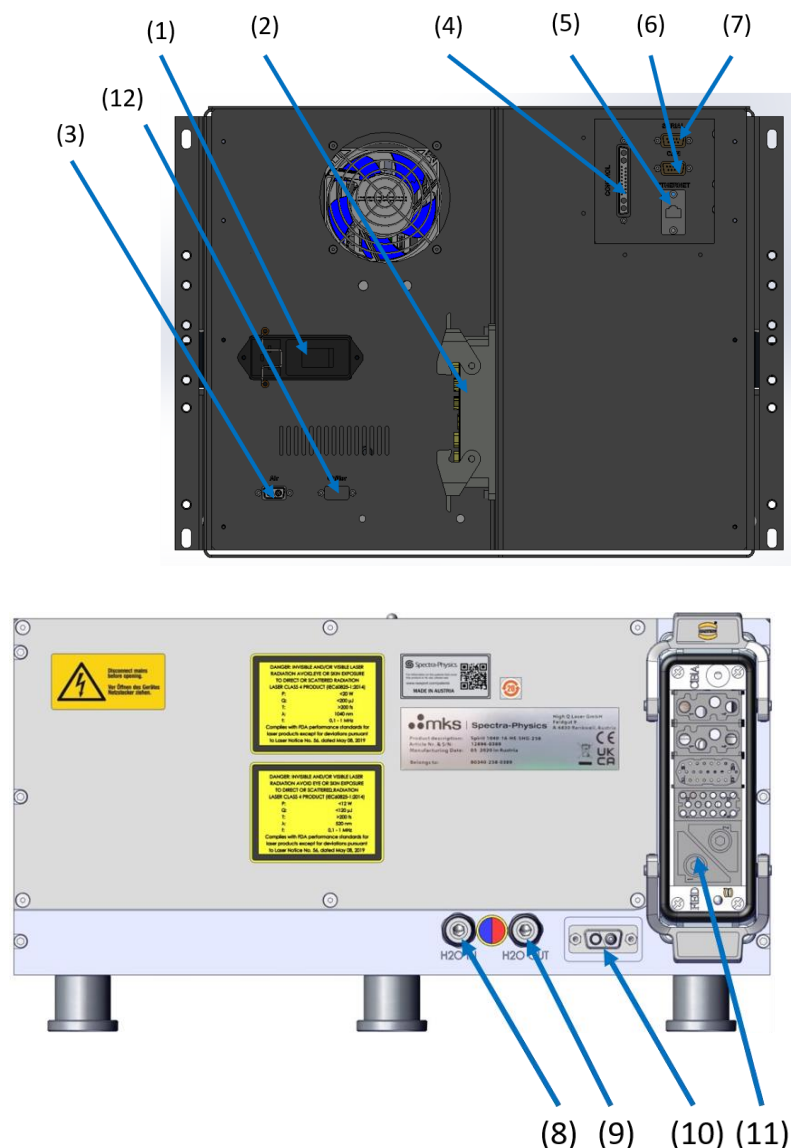
- 5) The laser system can either be controlled using the ethernet port **(5)**, the Can-Interface **(6)** or the serial port (optional)¹ **(7)**.

To use

- the ethernet port attach a standard ethernet cable.
- the Can-Interface attach the delivered standard serial cable and connect it to the IXXAT-Adapter and the host computer.
- the serial port attaches a standard serial cable.

5.2.4.2 SPLIT Rack (Optional)

Before setting up cable connections between electronics rack and the laser head, ensure proper grounding of the electronics rack through the main plug **(1)**. Do NOT switch on the electronic rack using the main switch (I / O) until all cables and water hoses are connected.



- 1) Attach the power cable to the electronic rack **(1)** and plug the line cord into a grounded outlet.
- 2) Attach the control cable between the electronic rack **(2)** and the laser head **(11)**.

¹ For more detailed information please contact Spectra-Physics.

- 3) If the delivered Spirit® laser system is equipped with an optional SHG compartment, the integrated SHG box has to be flushed with purified air. Therefore, attach the air purifying cable between electronic rack **(3)** and laser head **(10)**.
- 4) The user control interface has to be attached to the electronic rack **(4)**. The connector plug configuration is described in APPENDIX A3: User Control Interface.
- 5) The laser system can either be controlled using the ethernet port **(5)**, the Can-Interface **(6)** or the serial port (optional)² **(7)**.

To use

- the ethernet port attach a standard ethernet cable.
 - the Can-Interface attach the delivered standard serial cable and connect it to the IXXAT-Adapter and the host computer.
 - the serial port attaches a standard serial cable.
- 6) Attach the control cable between the electronic rack **(12)** and the chiller system.

Caution:

Use only the provided power cord or a power cord supplied by Spectra-Physics or a representative. Ensure proper grounding/earthing of the power inlet. Please note that the mains power supply requirements must be met according to local regulations.



5.2.5 Installation Checklist

Verify all of the following items:

- The laser and electronic rack are mounted securely in place.
- The chiller is installed with proper ventilation and filled with additive or distilled water.
- The power cable is attached to the electronic rack and plugged into a grounded outlet.
- The control cable is installed between electronic rack and laser head.
- If installed, the air purging tubes are securely connected, and hoses kept clear of kinking or pinching.
- The user control interface is attached.
- If installed, the shutter relays and the interlock are properly wired.
- The graphic user interface (GUI) and driver are installed on the host computer.
- The laser head is connected to the host computer via either the serial cable or the Ethernet cable.
- Verify the output aperture is not covered or blocked.
- The mechanical laser safety shutter is opened.
- The laser beam, when present, is properly directed to terminate into an appropriate beam dump or power meter.

The system is now ready for use. Remove any protective covering from the laser's shutter or output window and place a suitable power detector or beam dump in front of the laser output port.

² For more detailed information please contact Spectra-Physics.

6 Communication

The Spirit® laser system can be controlled using the CANopen interface, or it can be controlled via the TCP/IP interface. The CANopen interface can be used in conjunction with the graphical user interface (GUI) control software that is included with the system (HMT) or by another CANopen master.

The communication interfaces provide an easy way to monitor and control the laser system. A description for system operation with the GUI control software can be found in chapter 7 “Operation”.

6.1 Connecting the Controller to a CANopen Network

6.1.1 CANopen protocols

The Spirit® laser system can be controlled using the CANopen interface. The following CANopen protocols are fully supported:

- SDO protocol
- PDO protocol
- NMT protocol.

Please note, that the following CANopen protocols are not implemented:

- SYNC protocol
- EMCY protocol
- Timestamp protocol
- Heartbeat protocol
- Bootup protocol
- Node guard protocol.

6.1.2 CAN In, CAN Out

The controller provides one connector for communication over a CANopen network labeled with **CAN**. The interface is galvanically isolated.

With this connector it is possible to connect the controller as an end device with included termination of the CANopen bus.

The **CAN** connector is a female 9 pin D SUB connector at the back of the SPIRIT Controller. This connector has to be connected to the CANopen master. The pin assignment can be found in the table below:

Pin	Signal	Description
2	CAN_L	CAN_L bus line (dominant low)
3	CAN_GND	CAN ground
7	CAN_H	CAN_H bus line (dominant high)

All other pins are not used. 120 Ω bus termination resistors are present inside the controller between **CAN_LOW** and **CAN_HIGH** to ensure proper operation of the CANopen bus.

6.1.3 Node ID

Every CAN Open Device must have a unique node ID. The node ID of the SPIRIT Controller is set to 0x0E (fixed).

6.1.4 Baud rate

The default **baud rate** is **125kBit/s** and can be set up to 1MBit/s via a CAN command (see 11 Appendix A1: Spirit CANopen Communication Overview). Changing the baud rate requires cycling the power of the SPIRIT Controller.

6.2 Using the CANopen interface

Please see CIA Draft Standard 301: Application Layer and Communication Profile.

6.2.1 Network Management – Application State Machine

The application state machine is coupled to the NMT (Network Management) state machine. Operation of the laser is only possible after sending the Start Node indication.

6.2.2 User / Service / Factory Profiles

Three different communication profiles are used. The user profile allows the customer to control the laser system. All other profiles are password protected (see 11 Appendix A1: Spirit CANopen Communication Overview for available parameters).

6.2.3 Control, PDO Mapping

The NMT state machine has to be in the state **Started**. The necessary parameters to control the laser are mapped into **Process Data Objects (PDO)**. The PDOs are mapped statically, are asynchronous by default and the TxPDO is event triggered.

Name	COB ID	Typ	Transmission Mode	Trigger Mode	Mapped Objects				
					Databyte 0	1	2	3	4
RxPDO1	200h + Module ID	Receive			Control (Object 2000h)				
TxPDO1	180h + Module ID	Transmit	Asynchronous	Event Driven	Status (Object 2001h)	Common Warnings (Object 2100h)		Common Errors (Object 2101h)	

Use the **Control parameter** to change the state of the laser. The actual state is stored in the **Status parameter**. The warning and error register uses the same codes. In case of an **error** the system will automatically switch to the **error state** and remain there until a new control command is sent.

6.2.4 SDO Parameter (Service Data Objects)

With the low priority SDO parameter the possibility is given to access the controller's **Object Directory (OD)**. With the **Command Specifier** the access mode can be chosen (read/write).

Name	COB ID	Typ	Transmission Mode	Trigger Mode	Mapped Objects				
					Databyte 0	1	2	3	4
SDO Master	600h + Module ID	Receive			Command Specifier	OD Index		Subindex	
SDO Slave	580h + Module ID	Transmit	Asynchronous	Event Driven	Command Specifier	OD Index		Subindex	

6.2.5 Error Handling via EMC Message

If an **Error** occurs, the controller will switch to the **Error state**. Operation of the controller will be stopped, also the measurement of active values. This means that in the **Error state** all parameters contain the latest valid values making a further debugging easier. The controller will remain in this state until a new control command “write” is performed (see state machine).

Also, TxPDO1 is sent, including the Common Error and Common Warning codes. In addition, an emergency message is sent. The bytes 3 to 6 contain detailed information on the error. Depending on the value of the Common Error register (bytes 3 and 4) an Error code belonging to the affected sub module is sent (bytes 5 and 6). See 9.1 “Error Codes” for all possible Error Codes.

Byte	0	1	2	3	4	5	6	7
Description	Emergency Error Code		Error Register (Object1001h)	Manufacturer specific Error Code				
Parameter				Common Errors (Object 2101h)		Error Register depending on Common Errors: see 11 Appendix A1: Spirit CANopen Communication Overview		not used

6.3 Using the TCP/IP interface

6.3.1 Overview

This manual describes the Spectra-Physics Spirit® command interface using TCP/IP. The Spirit® laser system is divided into several sub-devices, for example the oscillator, amplifier and chiller. Each sub-device has its own warning and error parameters.

Network Protocol	TCP
Default Port	9000
Case Sensitive	Yes
Allowed Line Separators	\r, \n, \r\n, \n\r (\r: Carriage Return ASCII 13, \n: Line Feed ASCII 10)
Allowed Symbol Separators	' , \t (' : Space ASCII 32, \t: Horizontal Tab ASCII 9)

6.3.2 Command

Query Parameters

Request (sent from Client (e.g. PC) to Host (Laser)):

```
PARAMETER__NAME [LINE END]
```

Response if request is successful (from Host to Client):

```
OK VALUE PARAMETER__NAME\r\n
```

Response if request fails (from Host to Client):

```
ERROR ERROR__NUMBER PARAMETER__NAME\r\n
```

Responses from Host to Client are \r\n terminated as line separator.

Example

The following example is requesting the value of `power_actual`:

```
power_actual\n
OK 3.12 power_actual\r\n
```

In this example `power_actual` is misspelled:

```
power_atcual\n
ERROR 1 power_atcual\r\n
```

Error number 1 means „Parameter not available“ (see “*Communication Error Numbers*”)

Control Parameters

Request (sent from Client to Host):

```
PARAMETER_NAME=VALUE [LINE END]
```

Response if request is successful:

```
OK PARAMETER_NAME\r\n
```

Response if request fails:

```
ERROR ERROR_NUMBER PARAMETER_NAME\r\n
```

Example

The following example is setting the value of `power_setpoint`:

```
power_setpoint=3.5\n
OK power_setpoint\r\n
```

Communication Error Numbers

Number	Description
1	Parameter not available
2	Parameter not writable
3	Format error in request
4	Value is out of range
5	Host is busy
6	Host is in wrong mode
8	Parameter not readable
12	Internal Timeout

6.3.3 Interface Settings

Information about the TCP/IP address are seen in the **Advanced Mode** -> **IP Configuration**. The actual IP address is shown under **Actual Value**. Two different modes are available for setting/getting the IP address. These modes can be chosen with a drop down menu. The default value set by the factory is “**Dynamic**”.

By choosing

- **“Dynamic”** the IP address will automatically be given by the DHCP server.
- **“Static”** the IP address is given by the user. Therefore, the required IP address is set under **Set Value** by *Static IP (0)*, *Static IP (1)*, *Static IP (2)* and *Static IP (3)*. The sub netmask is chosen under *Static IP Netmask (CIDR)*.

The **“Service”** Mode is not applicable for the customer and needed for the Service Engineer of Spectra Physics.

The new IP address or the changed mode will be set after power cycle of the rack.

Static Dynamic Service

Set IP Address

Actual IP Address

Set Value :	
Name	Value
Mode	Dynamic
Static IP [0]	10
Static IP [1]	0
Static IP [2]	0
Static IP [3]	21
Static IP Netmask (CIDR)	8

Actual Value :	
Name	Value
Actual IP [0]	192
Actual IP [1]	168
Actual IP [2]	120
Actual IP [3]	86

7 Operation

This chapter illustrates how to operate the Spirit® laser system. The laser system can be operated either with the Spirit® GUI or via Ethernet TCP/IP. This section details the GUI controls. A full list of commands and queries is provided in chapter 11 “Appendix A1: Spirit CANopen Communication Overview” and chapter 12 “Appendix A2: Spirit TCP/IP Communication Overview”.

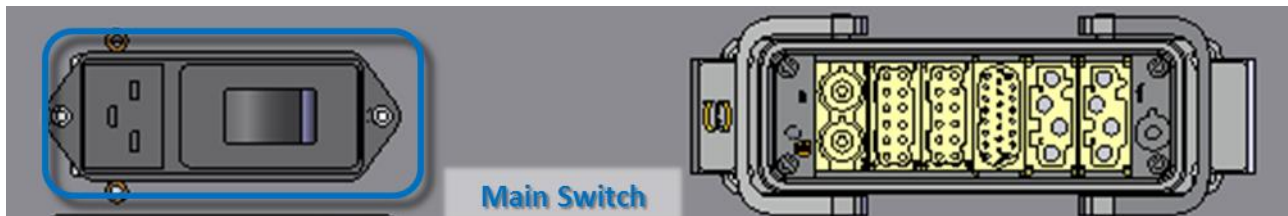
Warning:

Anyone operating or testing the laser needs to be trained on the relevant laser safety procedures.



7.1 Operating the laser system

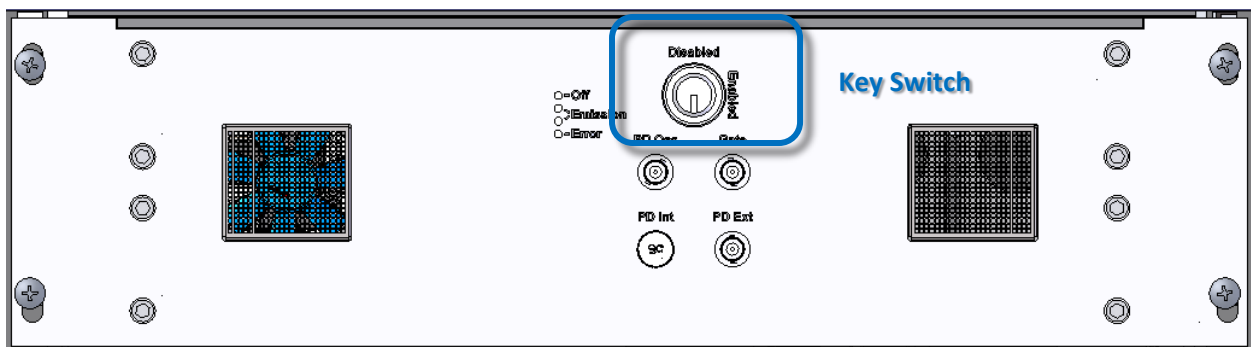
To put the system into operation set the main switch at the back of the electronic rack (I / 0) to I.



The key switch on the front of the power supply controller is intended for enabling / disabling the operation of the laser system.

The key switch must be set to **Enable** to ensure operation of the laser system. The key switch is NOT intended to activate or deactivate lasing of the system (turn-key system).

If the key switch is set to **Disable** during operation of the system (System Standby or System ON, including shutter open), the state will be switched to System OFF.



Caution:

To control the laser system the key switch must be in the **Enabled** position. The initial position of the key switch is **Disabled** in which the key can also be removed.

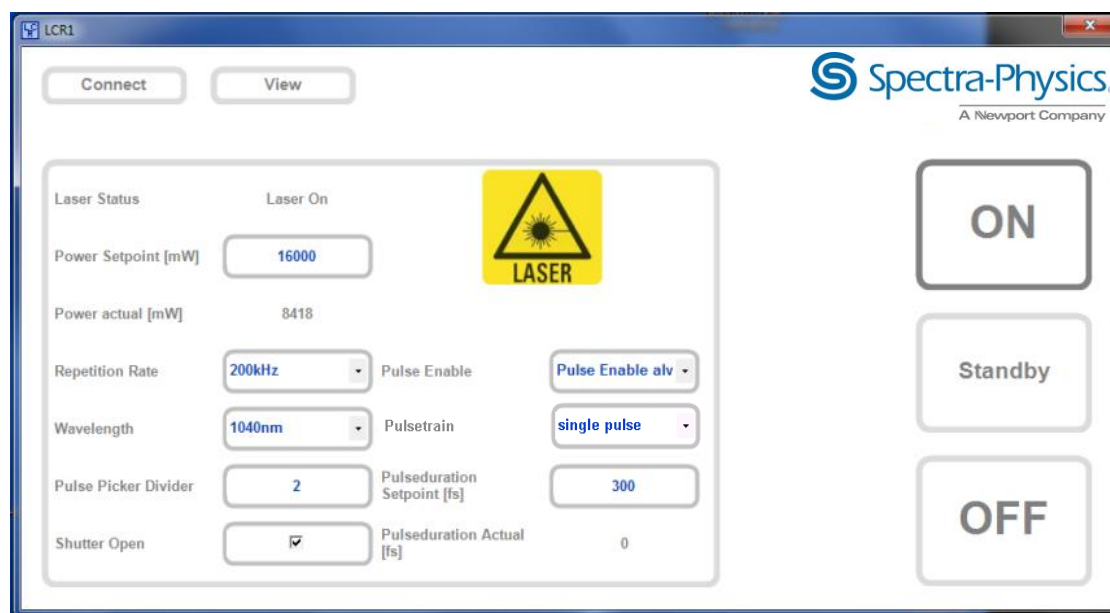


7.2 Graphical User Interface - HMT

Executing the provided HMT program will bring up the User Interface – shown below.

Press the “Connect” button to establish communication between software and the laser system. Once the connection is established, it is ready to use!

The following section describes the functionality which is provided with the main control panel.



Feature	Description
Button Connect	Press to connect the HMT to the laser system.
Button View	Press for Advanced View Operation.
Button ON	Press to switch the laser system to ON.
Button Standby	Press to switch the laser system to Standby.
Button OFF	Press to switch the laser system to OFF.
Laser Status	Displays the laser status.
Power Setpoint [mW]	Set the laser output power in mW.
Power actual [mW]	Displays laser output power in mW.
Repetition Rate	Sets the repetition rate.
Wavelength	Sets the wavelength to 1040nm or 520 nm (optional).
Pulse Picker Divider	Sets the Pulse Picker Divider.
Shutter Open	Opens or closes the electric-mechanical shutter.
Pulse Enable	Set Pulse Enable Mode: 0: External; 1: Pulse Enable always on; 2: Differential. Basic Setting 1: Pulse Enable always on.
Analog IN	Set Analog IN to disabled or enabled.
Pulse Duration Setpoint [fs]	Set the pulse duration in fs (optional).

7.2.1 “System OFF”

The initial status of the laser system after switching on the main power is **System OFF**. In the status **System OFF** no laser radiation is emitted.

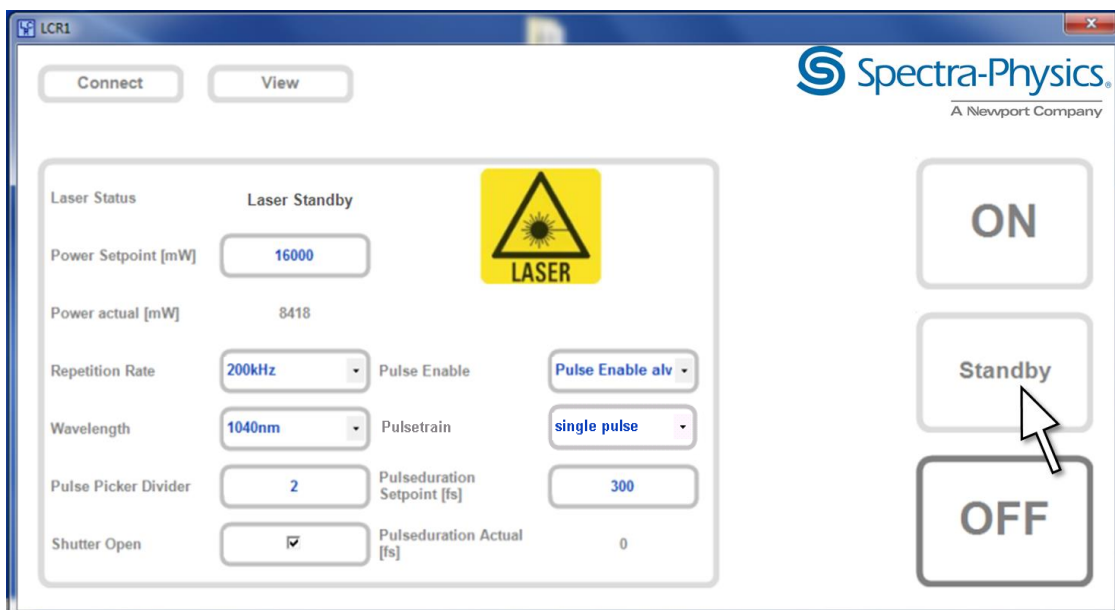
Information:

Set the operation mode **System OFF** when you are not working with the laser system.



7.2.2 “Standby”

Press the **Standby** button to change from the status **Laser Off/Laser ON** into Standby. In **Standby**, the temperatures of the laser head are still stabilized, and the laser diode current of the amplifier and the seeder are on. The HV Switch is off, so that no amplification takes place. The electro-mechanical shutter is closed.



Information:

The operation mode **Standby** can be used when the user stops working with the system for a short period of time. In the **Standby** mode the laser stays stabilized, and the further work can be continued within a few seconds.



Important:

When the laser is warming up from a cold-start, allow 30 minutes for the laser to completely stabilize.

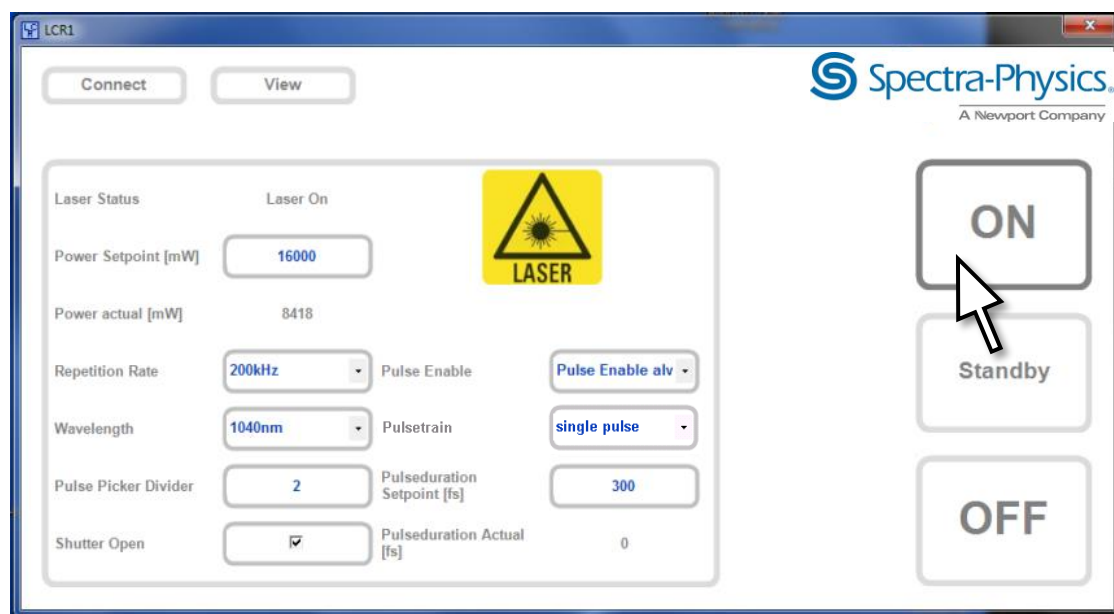


7.2.3 “System ON”

Press the button **ON** for putting the laser into **ON** status. The laser system is now ready for use. To start laser emission press **Shutter Open**.

Caution:

The Spirit® laser is a Class IV—High-Power Laser whose beam is, by definition, a safety and fire hazard. Take precautions to prevent accidental exposure to both direct and reflected beams. Diffused as well as specular beam reflections can cause severe eye or skin damage.



Important:

Always verify that the laser output beam is properly terminated before switching the laser into the **ON** state.



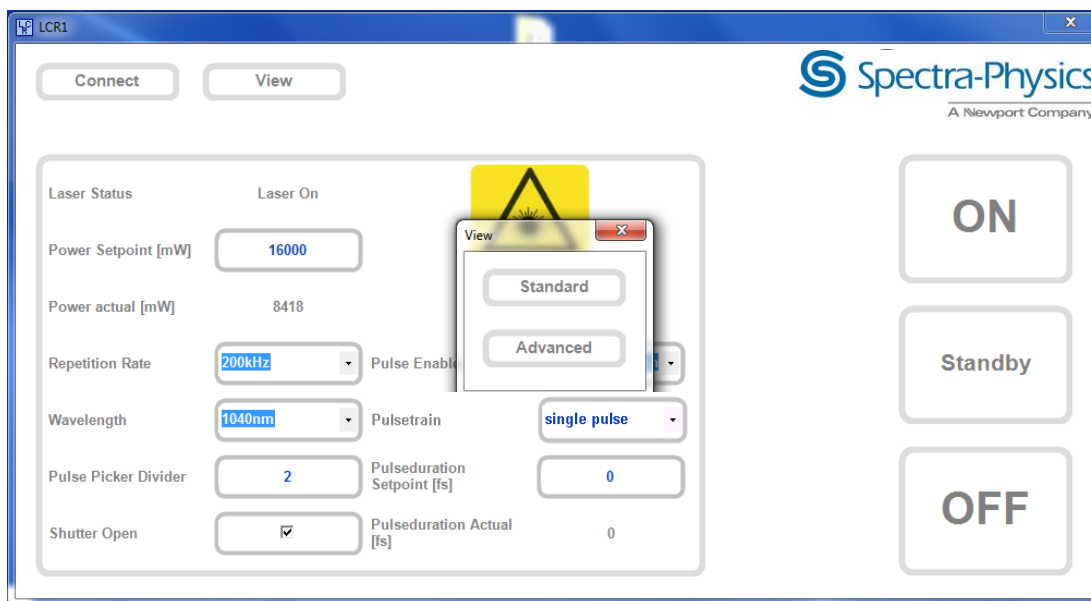
Information:

To allow laser emission it is necessary to ensure that in addition to the electro-mechanical shutter, the mechanical shutter is also opened. The mechanical shutter is located at the front panel of the laser can only be opened manually (see chapter 5.2.1.)



7.2.4 “Advanced View”

Pressing the **View** button offers a choice between **Standard** or **Advanced**. The **Standard** view is the normal user interface. The **Advanced** view shows detailed information about the laser system, for example system-internal temperatures, actual currents and several other parameters. In addition, detailed information about errors and warnings can be found here in case the system goes into an **Error** state.



7.2.5 Shutter Control and Laser Safety

In the operation mode **System ON** the shutter can be closed and opened using the **Shutter open** check box.

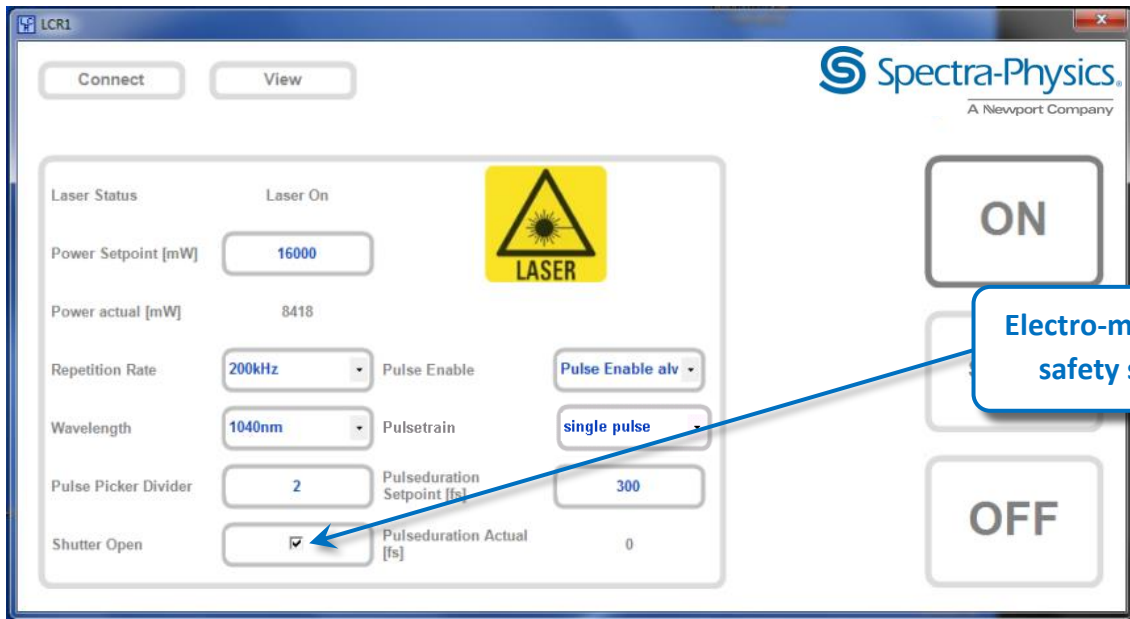
If the laser is set to **On** from **Standby** or **Off** mode, the electro-mechanical shutter needs to be opened, as the shutter is automatically closed in **Standby** or **Off** mode.

In case of an error the electro-mechanical shutter is closed immediately. If the interlock loop is opened the shutter is forced to close. The shutter state is continuously observed by the Spirit® controller.

Caution:

The integrated shutter is a **safety shutter** and **must not** be used as a **process shutter** or as a **beam block**.





7.2.6 Repetition Rate Adjustment

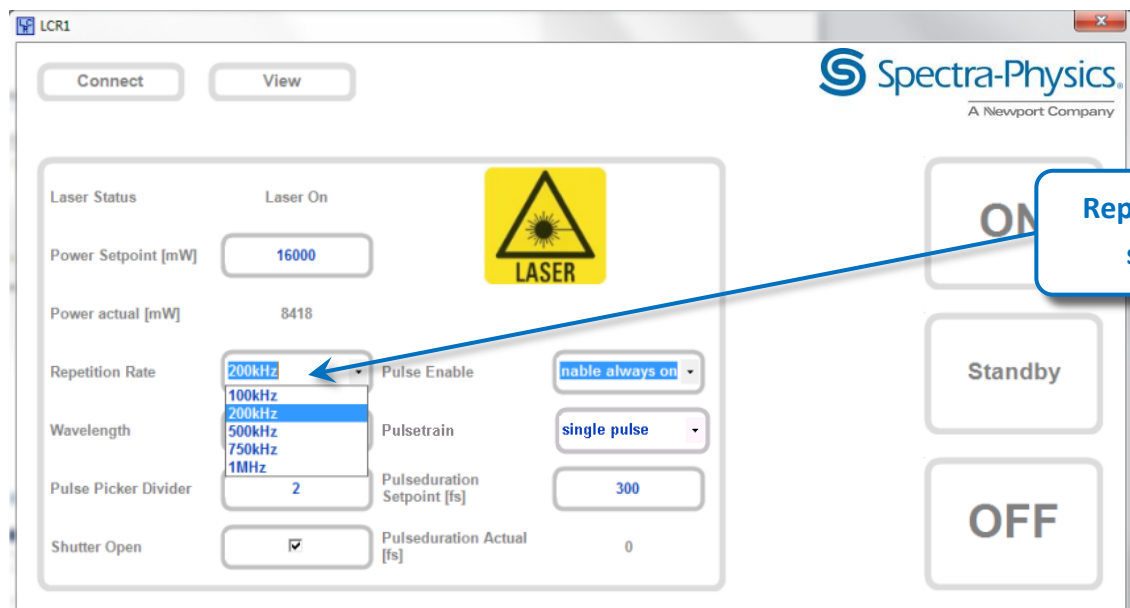
On the front panel of the graphical user interface the repetition rate can be changed using the drop down menu. The repetition rates are pre-defined and calibrated in the factory.

The following repetition rates are available for the SPIRIT 1040-16 HE:

- 100 kHz
- 200 kHz
- 500 kHz
- 750 kHz
- 1000 kHz (1 MHz)

For the SPIRIT 1040-30 the available repetition rates are:

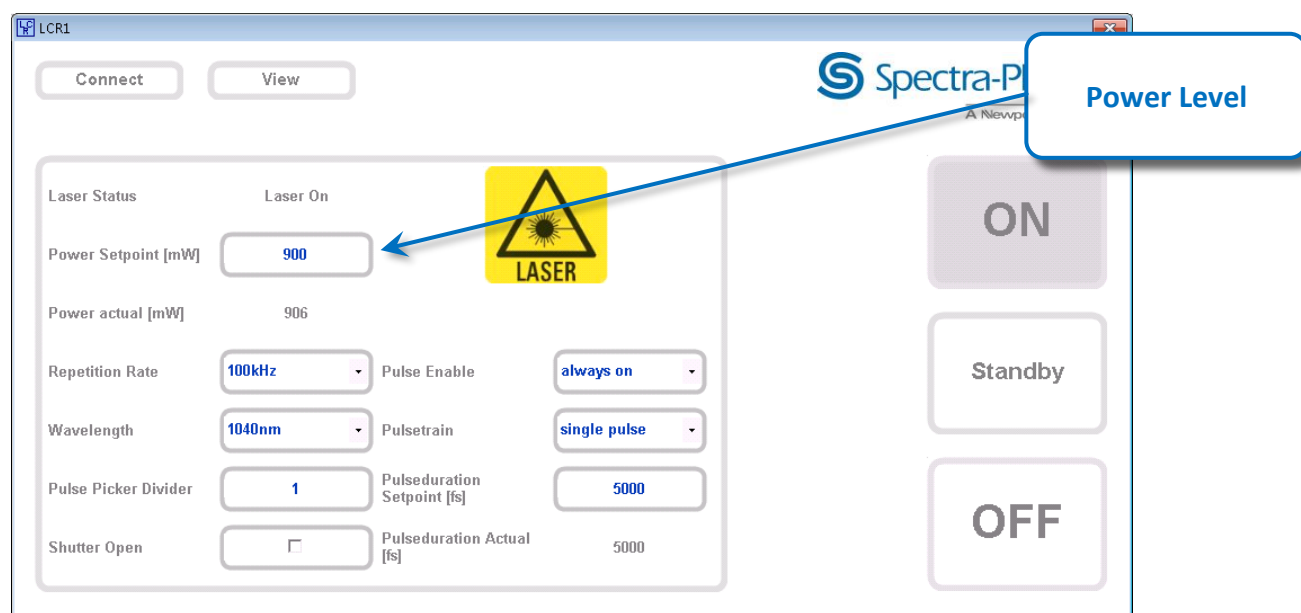
- 250 kHz
- 500 kHz
- 750 kHz
- 1000 kHz (1 MHz)



The system's highest pulse energy is available at a repetition rate of 100 kHz in case of the SPIRIT 1040-16 HE, and 250 kHz in case of the SPIRIT 1040-30 HE model. Repetition rates below 100 kHz/250 kHz (16 W / 30 W) are accessible via the internal pulse picker. Every basic repetition rate can be divided by any integer >0 up to 10^7 . For example, a repetition rate of 750 kHz with a pulse picker divider of 5 provides a repetition rate of $750 \text{ kHz} / 5 = \mathbf{150 \text{ kHz}}$. The pulse energy will be the same as at 750 kHz, but the output power is lowered to 1/5 (for example 3.2/6 W instead of 16/30 W).

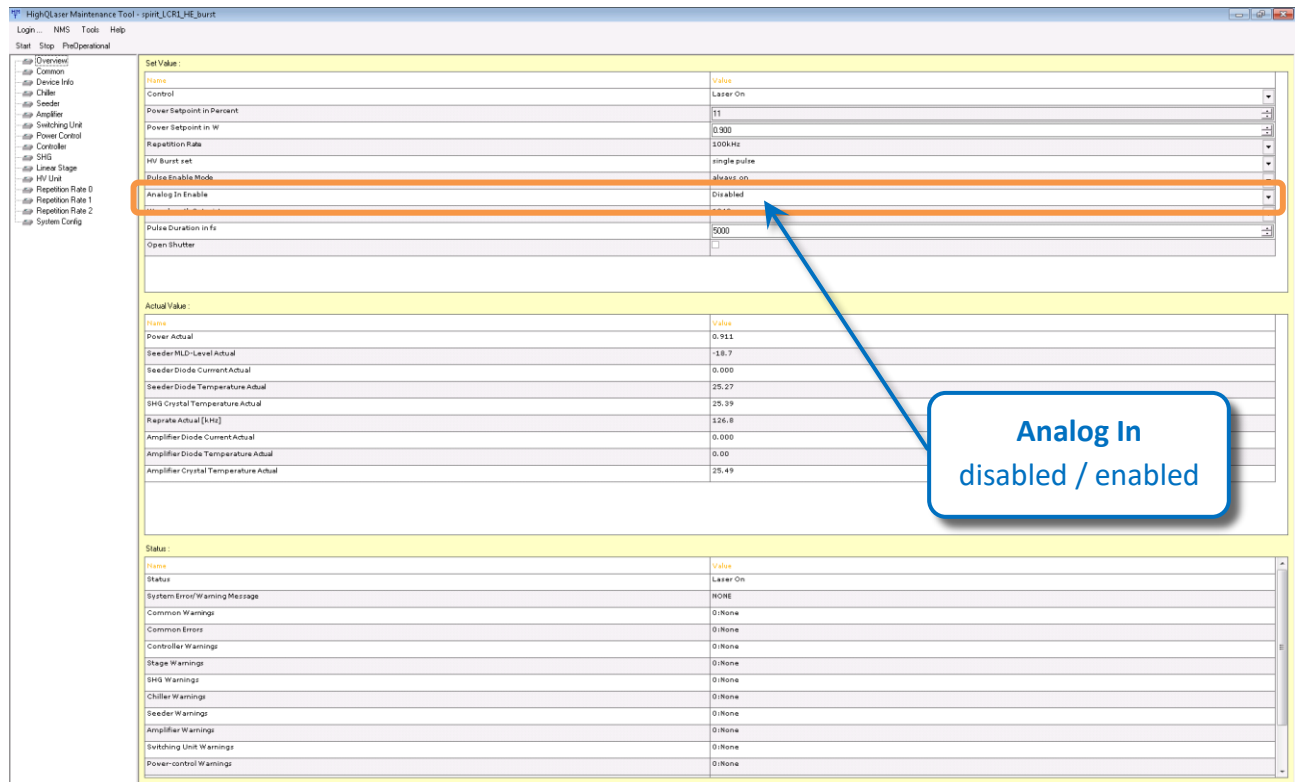
7.2.7 Power Control

The Spirit® output power and pulse energy can be controlled via the integrated pulse picker – either by software (“**Attenuator Setpoint / Power Level**”) or hardware (“**Analog In**”). To change the power control mode, one of the following settings in the GUI has to be selected:



- Control by software: (*Analog In set to Disabled*)
The Laser output power can be set by **Power Level** in mW. Additionally, all parameters can be changed in the advanced view (Power Control Submenu), too.
- Control by Analog In: (*Analog In set to Enabled*). In this mode the output power can only be controlled using an **external signal (Analog In)**, provided to the **User Control Sub-D connector** (see chapter 13 APPENDIX A3: User Control Interface). See also chapter 7.2.8 Laser Power Modulation using “Analog In”.

7.2.8 Laser Power Modulation using “Analog In”



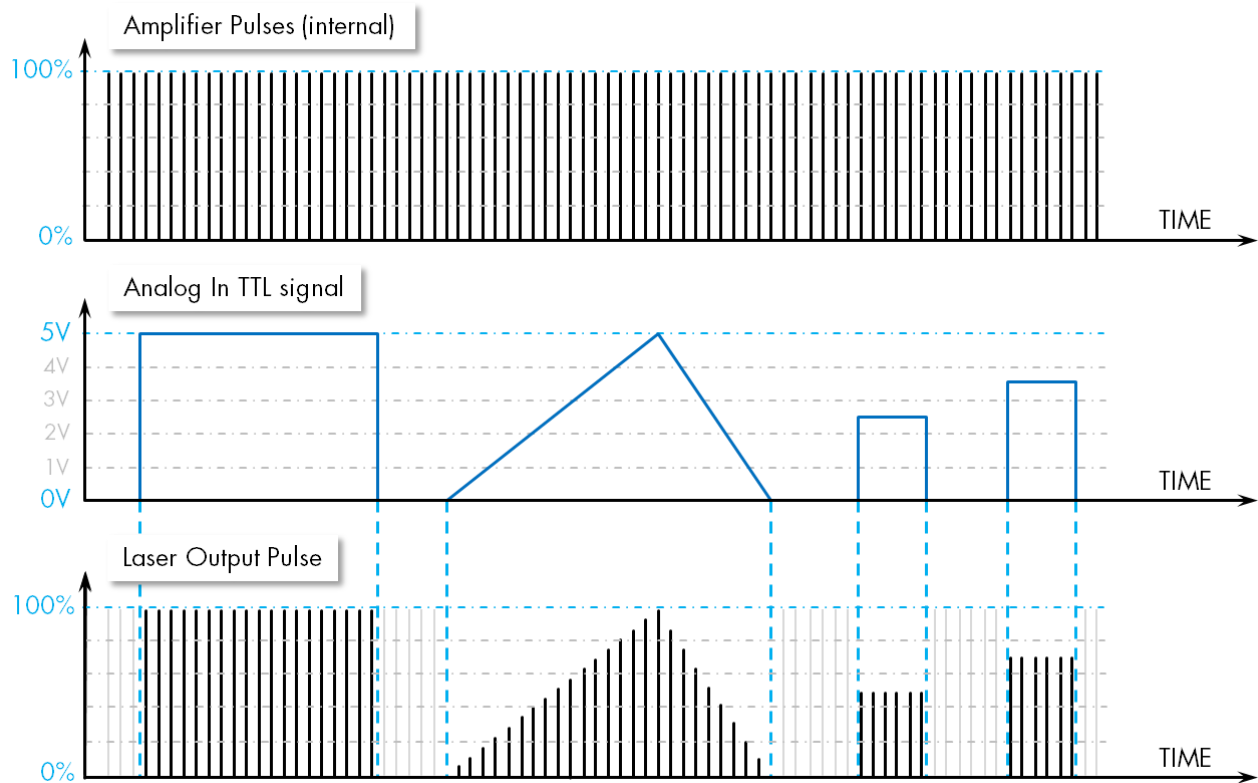
This mode of power control by an external signal is called **Analog In**. It can be enabled or disabled in the Advanced View mode of the control software GUI (as shown in the picture above). The input range has a maximum resolution of 4095 steps (12bit). This corresponds to a voltage range of 0 – 5 VDC.

The value for the input range can be set under “**Analog In Voltage Range**” in the submenu “**SystemConfig**”. This parameter is password protected.

The Analog In voltage input is linearized. The power level can be calculated by:

$$\text{Output Level} = P_{\text{out_max}} * V / V_{\text{max}}$$

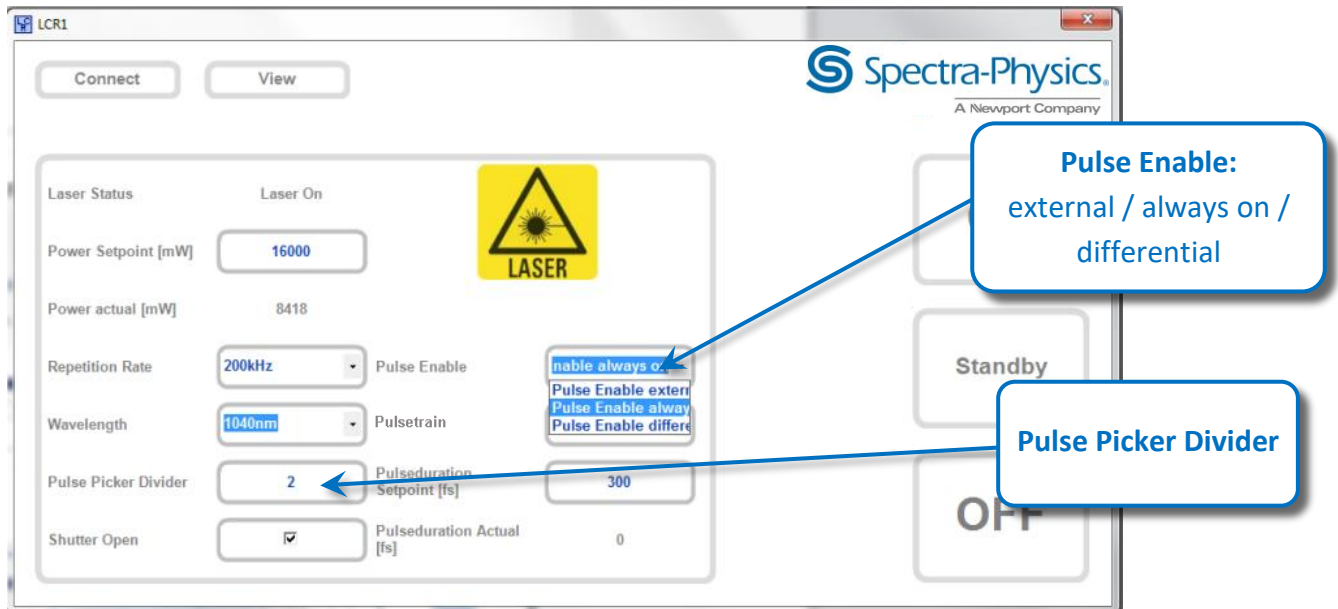
The Analog In feature offers a fast, easy-to-use and high flexible possibility for power and energy control and can be used to generate pulse trains on customer’s demands.



7.2.9 Pulse Enable and SYNC Out

The integrated pulse picker can be used to gate the pulse train using the **Pulse Enable** feature. This signal is a TTL input signal and can be accessed by the BNC connector at the User Control Interface (see APPENDIX A3: User Control Interface).

The BNC connector **SYNC Out** is a TTL output signal which has the same frequency as the set pulse repetition rate. This signal is synchronous with the pulse train and can be used to synchronize events to the pulse train from the laser.



Pulse Enable set to **Always on**:

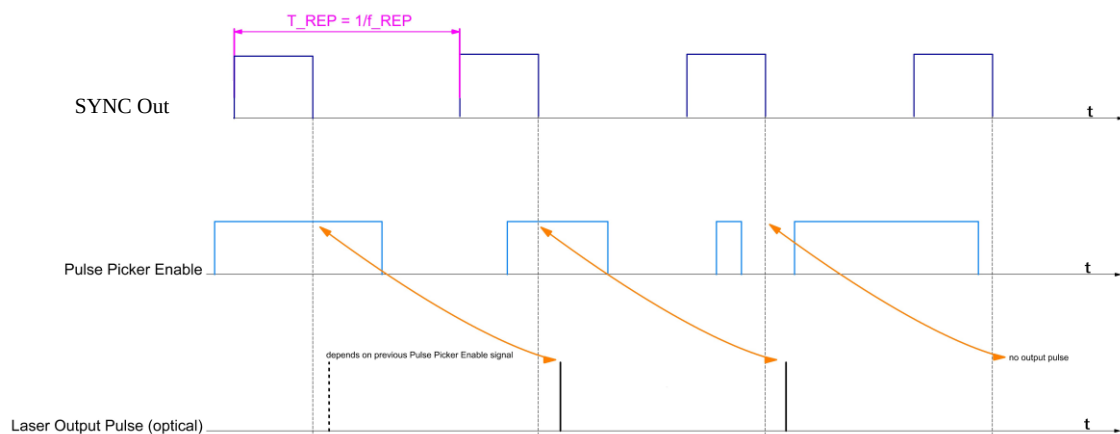
If Pulse Enable is set to **Always On**, no external 5V signal will be needed at the Pulse Enable input. Dividing the repetition rate by an integer divider is possible (output pulse train frequency = basic repetition rate / divider).

Pulse Enable set to **External**:

The external control of the pulse picker is activated, and the laser output is controlled by the external gate signal:

0V TTL input: Pulse Disabled = **pulse not emitted**

5V TTL input: Pulse Enabled = **pulse emitted**



The Pulse Enable signal is latched to the falling edge of the Sync Out signal (as shown in the picture above). If a 5V TTL signal is attached, the laser pulse of the next Regen Cycles is switched out. Dividing the basic repetition rate by an internal divisor is still possible in this mode.

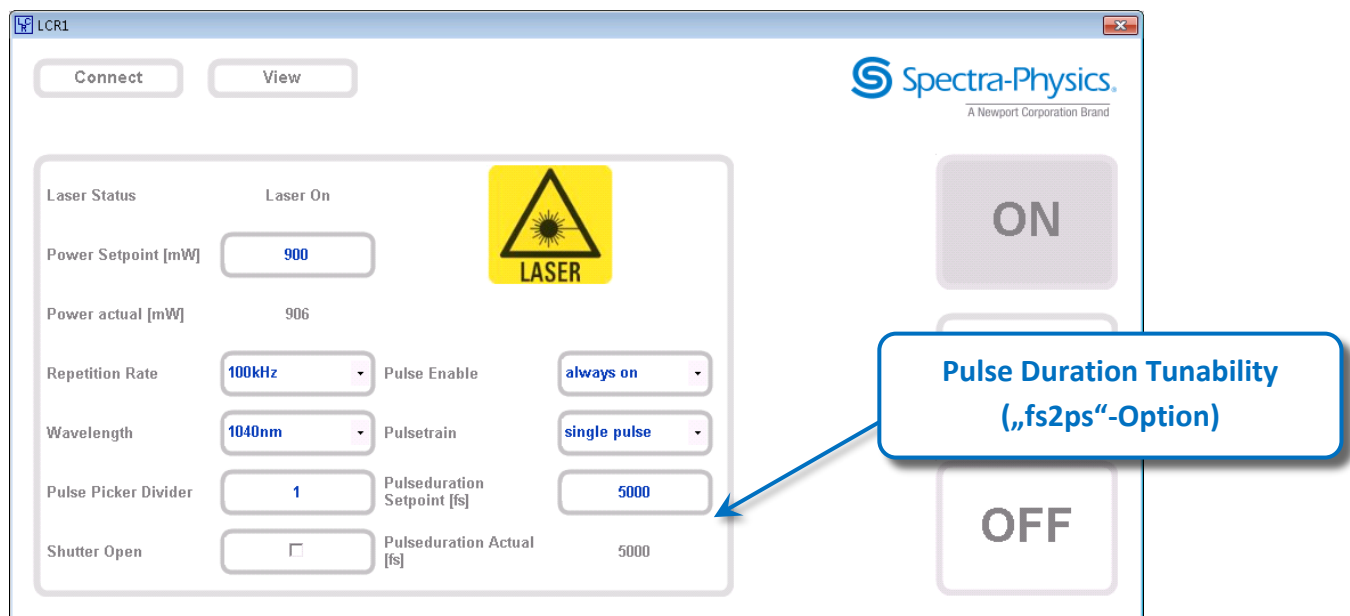
7.2.10 Variable Pulse Duration (optional)

The laser system has an integrated automatic pulse duration control (optional). The desired pulse duration τ_p (in units of fs) can be set between <400 fs and >10.0 ps. It is recommended to always double-check the pulse duration with an external auto-correlator.

Information:

The minimal achievable pulse duration is repetition rate dependent and can be found in the Quality Test Report (QTR) that accompanies this manual.

If the laser system can also be changed to 520nm in wavelength, please be aware that the maximum conversion efficiency is achieved at the minimum pulse duration and that changing the pulse duration leads to a loss in output power at 520 nm.



Information:

When entering a pulse duration value smaller than the shortest pulse duration (<400fs) or larger than the longest pulse duration (>10.0ps), the system will not change the pulse duration at all.



Information:

The pulse duration is altered by de-tuning the internal pulse compressor. Therefore, the pulses are not transform-limited but always have the same optical spectrum independent of the pulse duration chosen.



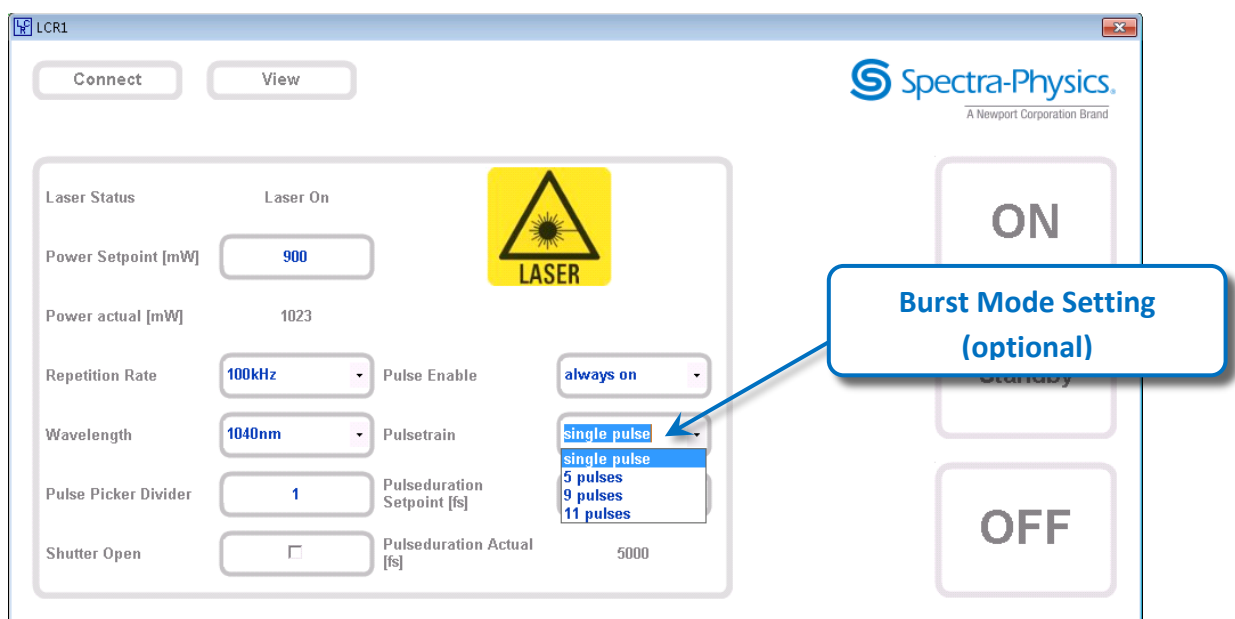
Information:

After a restart of the laser system by rebooting the control rack or the control software (GUI), the pulse duration will automatically be set to its last chosen value.



7.2.11 Burst Mode (optional)

For the Spirit® laser series a Burst Mode option is available. With this option the customer can choose between “single pulse” operation (as in the standard configuration) and “burst mode” operation, in which the single pulse is split into a certain number of sub pulses with a temporal separation of approx. 12 nsec. The pulse energy of each intra-burst pulse is a fraction of single pulse’s total energy. For example, a distribution of an 80 µJ pulse into 5 burst pulses will result in 16 µJ for each of the 5 burst pulses. The number of pulses per burst can be selected on GUI’s standard view as shown below.



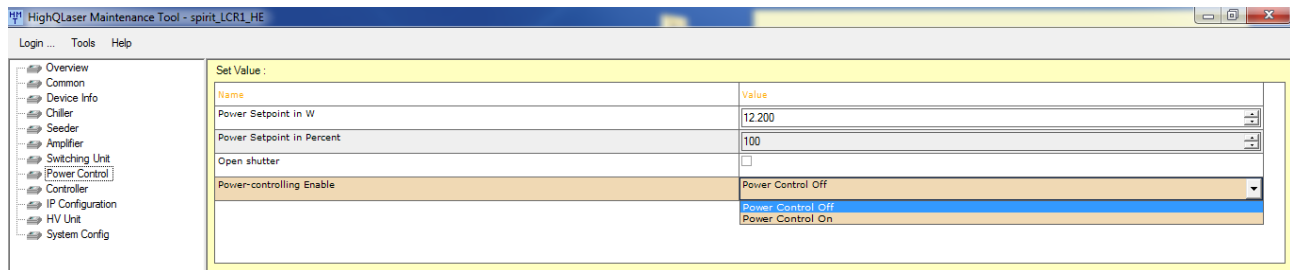
Information:

The conversion efficiency of the Second Harmonic Generation depends on the pulse energy at 1040nm and is optimized for the highest available pulse energy and single pulse operation. Therefore, the conversion efficiency will be lower in burst-mode operation.



7.2.12 Power Lock

The Spirit® control soft- and hardware is also equipped with a constant power mode, automatically stabilizing the output power to a selected value, called **Power Lock**. The Power Lock feature is based on a feedback control utilizing the integrated power sensor at the laser exit. The option Power Lock is by default deactivated. It can be activated in the Advanced View mode of the control software GUI (as shown in the following picture). It can be activated/deactivated under “**Power-controlling Enable**” in the submenu “**Power Control**”.



Information:

The Power Lock will be automatically deactivated, if the **Analog In** mode or the **Pulse Enable External** mode is used.



7.2.13 Auto Tune

The complete Spirit® laser system (electronic drivers, water chiller and optical components) is performing a self-check routine in the beginning of every cold-start process (electronic rack switched ON) to optimize system performance.

8 Maintenance and Service

The Spirit® laser head is designed for OEM applications and requires no alignment by the operator. Compensating for the aging of components or for changes in environmental conditions is possible during annual periodic maintenance.

This chapter describes the procedures used to maintain the output of the Spirit® and to extend its lifetime.

Caution:

The Spirit® laser is a Class IV—High-Power Laser whose beam is, by definition, a safety and fire hazard. Take precautions to prevent accidental exposure to both direct and reflected beams. Diffused as well as specular beam reflections can cause severe eye or skin damage.



Important:

Do not attempt to repair the system. Instead, report all problems to Spectra-Physics. If you experience any problems with this system or any equipment purchased from Spectra-Physics, or if you are in need of technical information or support, contact Spectra-Physics.



8.1 Maintenance

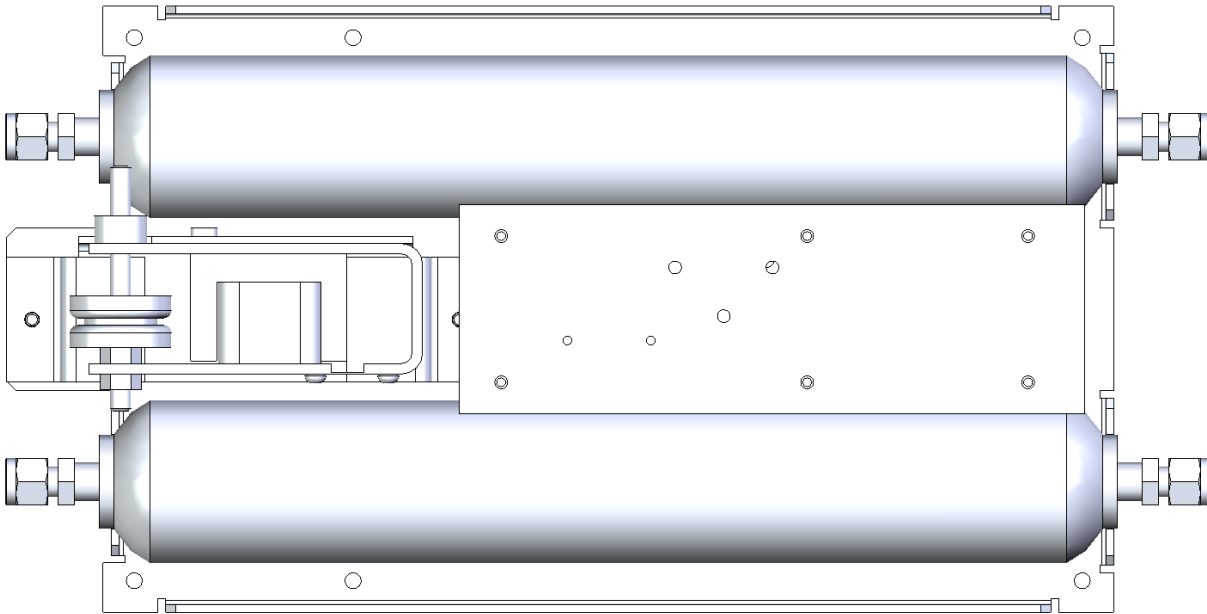
To ensure optimal and constant performance of the laser system Spectra-Physics recommends the following system check and pre-emptive maintenance steps:

Frequency	Action
Periodic maintenance items (by customer)	• Check the coolant level of the water cooling unit (chiller) monthly and refill the water tank, if necessary • Only dry cleaning of the product's surfaces (for example removal dust)
Six months (by customer /service engineer)	• Exchange chiller coolant (as recommended by manufacturer)
Annually (by customer/service engineer)	• Verification of software parameters • Exchange the Active Laser purge System (ALPS)
3-5 years / > 26,000 hours (by factory expert; not free of charge)	• Laser refurbishment in factory (recommended)

8.1.1 Changing the Active Laser Purge System (ALPS)

8.1.1.1 Stand-Alone Electronics Rack

The power supply rack of the Spirit 1040-16-HE-SHG laser system contains an **Active Laser Purge System (ALPS)**.



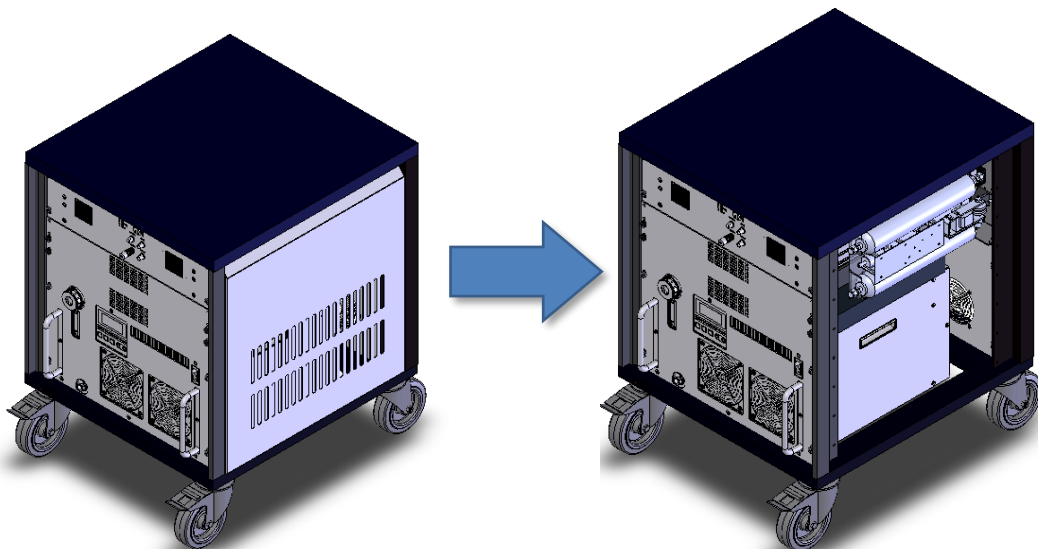
Important:

It is recommended that the ALPS module is replaced **annually**.

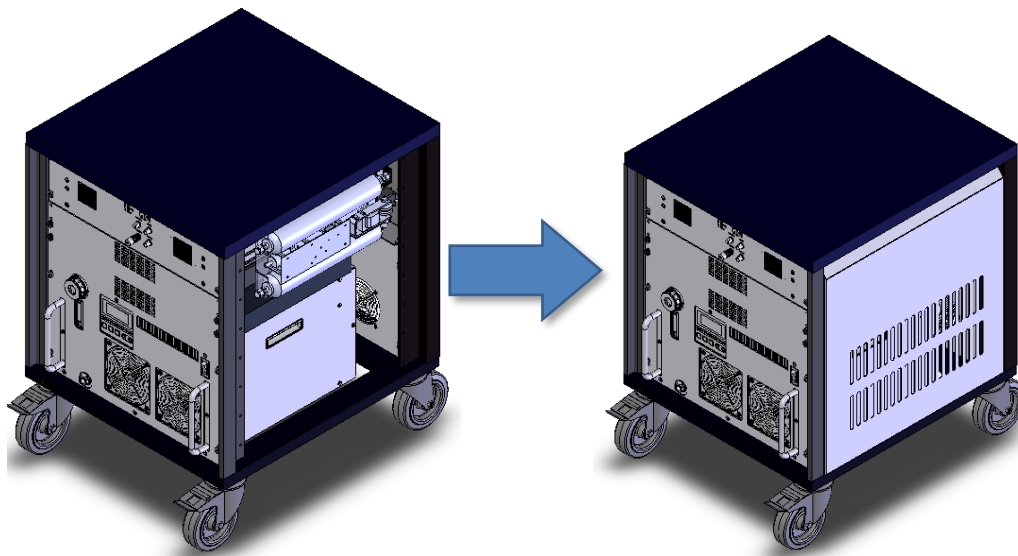
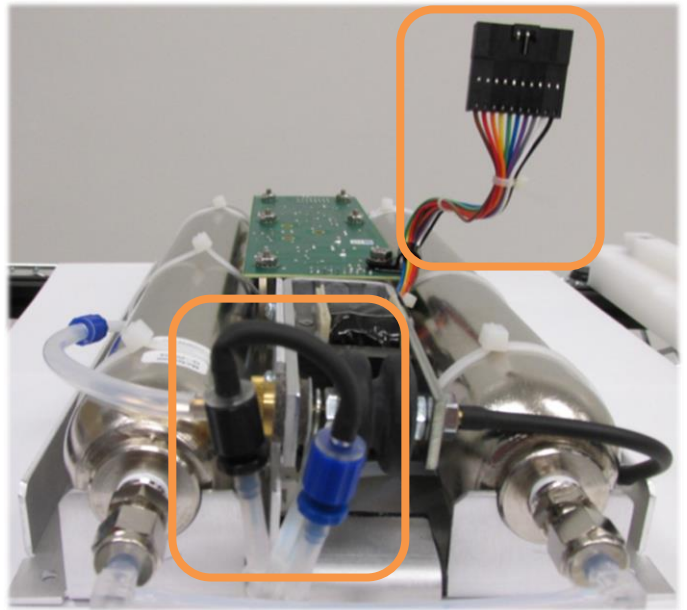


To exchange the ALPS module:

1. Turn off the AC power of the power supply rack and disconnect the AC line cord from the back of the rack.
2. Remove the side wall cover on the right side of the rack (looking on the rack's front side):



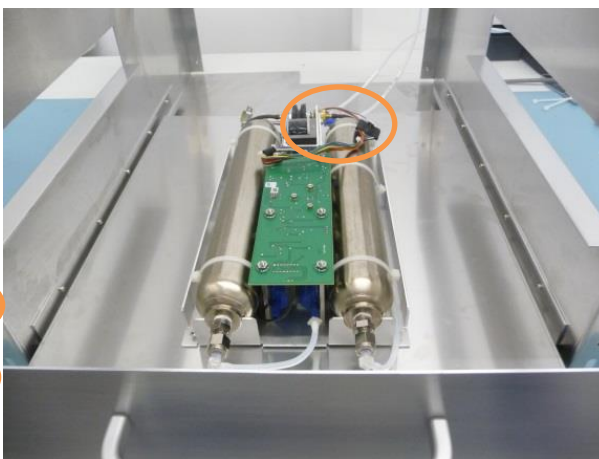
3. With the side wall removed, the ALPS module is now visible and accessible.
4. Disconnect the blue and black purge lines (note that they are rerouted) by turning the connector counter-clockwise until the line disengages.
5. Disconnect the multi-pin connector from the ALPS module.
6. Loosen the screw fixing the ALPS module on the Rack and remove the old module.
7. Attach a new ALPS module to the rack and tighten the screws.
8. Remove the purge line jumper from the new ALPS module and place it on the old unit.
9. Connect the blue and black purge lines and the multi-pin connector. To connect the lines, turn the connector clockwise until the line engages.
10. Attach the removed side wall to the rack and tighten the screws.



11. Plug the AC cord back into the rear of the rack and turn the AC power back on.
12. The Spirit laser system is ready for operation.

8.1.1.2 *SPLIT-Rack (OPTIONAL)*

1. Turn off the AC power of the power supply rack and disconnect the AC line cord from the back of the rack.
2. Remove the four screws fixing the drawer at the bottom part of the SPLIT rack.



3. The ALPS module is connected to the blue and black purge lines and to multi-pin connector. Therefore, pull carefully out the drawer.
4. Disconnect the blue and black purge lines (note that they are rerouted) by turning the connector counter-clockwise until the line disengages.
5. Disconnect the multi-pin connector from the ALPS module.
6. Loosen the screw fixing the ALPS module in the Rack and remove the old module.
7. Attach a new ALPS module to the rack and tighten the screws.
8. Remove the purge line jumper from the new ALPS module and place it on the old unit.
9. Connect the blue and black purge lines and the multi-pin connector. To connect the lines, turn the connector clockwise until the line engages.
10. Push carefully back the drawer. Fix the drawer with the four screws.

8.2 Service Contact

In case of malfunction or operation out of specification please get in contact with your regional Spectra-Physics sales or service contact person.

The addresses and contact data of the Spectra-Physics factory facilities are below:



A Newport Corporation Brand

Spectra-Physics Headquarter

1565 Barber Lane
Milpitas, CA 95035

United States

Tel +1 (408) 980 – 4300
Fax +1 (408) 980 – 6923

Email: service@spectra-physics.com
Web www.spectra-physics.com

Spectra Physics Rankweil

Feldgut 9
6830 Rankweil

Austria

Tel +43 (55 22) 826 46 – 100
Fax +43 (55 22) 826 46 – 800

Email: service@spectra-physics.at
Web www.spectra-physics.at

Information:

In case of any problem with the laser system, please call Spectra-Physics or one of the local representatives immediately to resolve the problem as soon as possible.



9 Diagnostics and Troubleshooting

This diagnostics guide is intended to assist in identifying some of the problems that may arise while using the Spirit® laser.

Important:

If there is a problem with the laser that you cannot resolve after using this basic diagnostics section, contact a Spectra-Physics service representative for further assistance.



9.1 Error Codes

In case of malfunction the laser / power supply will go to a **warning** or **error state**. The warning or error codes are either shown in the **Advanced View Mode** of the graphical user interface or can be read out via command through the **CAN interface** or **TCP/IP**.

A **warning state** will not stop the laser from running; the laser will still be emitting radiation in the System ON state.

An **error state** will set the laser to system OFF; the shutter will be closed, the diodes switched off and the HV supply set to zero. The laser can only be brought back to operation after the error has disappeared.

The possible warnings and errors are listed below:

Common

BIT	DEC	Description
0	1	Interlock (I-Lock)
1	2	Chiller
2	4	Seeder
3	8	Amplifier
4	16	SwitchingUnit
5	32	Power Control
6	64	Controller
7	128	System Error
8	256	Second Harmonic (SHG)
9	512	fs2ps Stage
13	8192	Master Off external
14	16384	Keyswitch
15	32768	Master Off internal

Chiller

BIT	DEC	Description
0	1	Temperature
1	2	Fluid Level Limit reached
2	4	Chiller Flow Rate

Seeder

BIT	DEC	Description
0	1	Diode Current
1	2	Diode Voltage
2	4	Diode Temperature
8	256	Modelock Signal out of specified range
9	512	Modelock Detector out of specified range

Amplifier

BIT	DEC	Description
0	1	Diode Current
1	2	Diode Voltage
2	4	Diode Temperature
5	32	Crystal Temperature

Switching Unit

BIT	DEC	Description
0	1	PDG Supply Voltage
1	2	PDG Interlock
2	4	PDG Seed Level
3	8	PDG Frequency
5	32	PDG Gate

Power Control

BIT	DEC	Description
0	1	Safety Shutter
3	8	Safety Shutter Relays Opened
4	16	Power Maximum Reached

Control

BIT	DEC	Description
0	1	Supply Voltage (24V)
1	2	Electronics Temperature LCR1
2	4	Amplifier Diode TEC Temperature
3	8	DSOSC Temperature
4	16	PSXTAL Temperature
5	32	HV Communication
6	64	Amplifier LD Driver Communication
8	256	Chiller Communication
9	512	DSOSC Communication
10	1024	Amplifier Diode TEC Communication
11	2048	PSXTAL Communication
12	4096	PDG Communication
13	8192	High Voltage Error
15	32768	SHG Attenuator Communication
16	65536	DSOSC Internal Error
17	131072	PSXTAL internal Error
18	262144	Amplifier LD Tec Internal Error
19	524288	Amplifier LD Driver Internal Error
20	1048576	fs2ps Stage Communication

Second Harmonic (SHG)

BIT	DEC	Description
0	1	XTAL Temperature
5	32	Airflushing System Filter

fs2ps Stage

BIT	DEC	Description
1	2	Stepper Not Calibrated
2	4	Stepper Moving
3	8	Stepper Internal Error
4	16	Stepper End Sensor Error

System-Error

No.		Description
0		No System Error present
2		Timeout initializing the system
3		Timeout reaching the initialized temperatures
4		Timeout initializing the Laser-Off state
5		Timeout initializing the Laser-Standby state
6		Timeout initializing the Laser-On state
7		Too many MQ-protocol-timeouts on the RS232-bus
8		LED error
10		More than 15 EEPROM write-cycles per minute
11		Configurator error
12		Data corruption of both copies (defaults loaded)
13		Data corruption of first copy (loaded from valid second copy)
14		Data corruption of second copy (loaded from valid first copy)
15		Update process error (defaults loaded)
16		Unsupported configurator
17		Data size differs (defaults loaded)
18		Configurator-update successfull
19		Memory access violation (read or write outside accessible memory area)
20		Corrupted data size
21		Too many HQ-protocol-timeouts on the default baudrate of 4800
22		Too many HQ-protocol-timeouts on the fast baudrate of 125000
23		Too many MQ-protocol-timeouts on the RS422-bus
24		Power-on reset
25		External reset
26		Brown-out reset
27		Watchdog reset
28		JTAG reset
29		Too many Modbus-protocol-timeouts (used for the SMC-chiller)
30		HVQ-Error
31		The HV-device temperature is out of range
32		HV-supply voltage error or input frequency error
33		Gate-signal did not go to 0 on time

9.2 Troubleshooting

Topic: Controller status LED “Error” is illuminated

The status LEDs on the controller’s front panel displays “Error”.

To quit the error, please change the laser status (laser off / standby / laser on) by using the software GUI or by sending a command.

If the error still occurs after changing the laser status, please restart the control rack. If the error appears continuously, please make a note of the error code (Advanced View Mode) and contact Spectra-Physics’ service department.

Topic: Laser switched off and controller/software show “error”

The laser control GUI or the status LEDs on the controller’s front panel show “Error” and the laser switched off.

To analyze the error, it is important to know the error code. To read out the error code, please do not change the laser state (on / standby / off) – this would delete the error code. Please go to Advanced View Mode in the software GUI and note the error code/message shown in the “Common” section. The error codes are described in chapter 9.1 “Error Codes”. Please call Spectra-Physics or one of the local representatives immediately to resolve the problem as soon as possible.

Topic: Analog In level shows value >0V

The software GUI shows an Analog In level of >0V (e.g. 1.85V), although no voltage is applied or no function generator is connected to the BNC connector of the customer control cable.

Please shortcut or terminate the BNC plug of Analog In to bring the level to 0V. The existing voltage value is caused by a floating grounding. If the Analog In feature is not needed, please chose “Analog In = disabled” to control the output power by software.

Topic: IR + SHG simultaneous output

The laser system can only be set to either xx% at 1040nm **or** xx% at 520nm. A combination of both wavelength at the same time cannot be set by the user.

Topic: >5 min to change laser states (Off / Standby / On)

To change the laser states from **Laser Off** to **Laser Standby** takes typically <5 minutes. The change from **Laser Standby** to **Laser On** is typically <1 minute. For initialization times >5 minutes, please go to: **GUI → Advanced View → Common**. Please check the warning or error messages/codes under **Warning List**. Please go to the corresponding sub module tab and take a screenshot to share the error message/code. Long waiting times are mainly caused by thermal problems (e.g. chiller temperature not stable → check hosing connections).

Please share the screenshot and a failure description with service@spectra-physics.at for analysis and support.

Topic: Power consumption

During the cold start of the laser system (max. consumption by chiller) the approximate power consumption is <1,500 W. In a levelled-off state the power consumption is approx. 1 kW.

Topic: Software installation for own PC

The Spirit®'s control software (GUI) can be installed on your own PC or laptop. Spectra-Physics provides a software installation package for Windows 32 Bit and 64 Bit systems. Please contact service@spectra-physics.at for further information.

Topic: Software compatibility with Windows versions

The Windows version requirements for Spirit®'s control software/GUI:

- Win XP, Win 7, Win 8 and Win 10
- 32 Bit and 64 Bit systems

If you want to use the software with another operating system, please contact service@spectra-physics.at.

Topic: Storage of shipment boxes

We highly recommend keeping and storing the shipment box of laser head and controlling rack. In case of a necessary factory repair or upgrade the safest way of shipment is in the original shipping crates. Otherwise new shipment boxes need to be ordered and sent to the facility first.

Topic: Basic Repetition Rates

The Spirit® laser system can be configured with up to 5 basic repetition rates: 100kHz, 200kHz, 500kHz, 750kHz and 1MHz. For additional specific repetition rates, please contact sales@spectra-physics.at or service@spectra-physics.at for possible upgrades or modifications.

Also, the integrated pulse picker (AOM) can be used to select repetition rates below the basic repetition rates, but always **integer fractions of the selected basic repetition rates** (e.g. basic repetition rate = 200 kHz → pulse picked repetition rates = 200 kHz, 100 kHz, 66.7 kHz, 50 kHz, 40 kHz, ...). By using the pulse picker the pulse energy stays constant and the output power is reduced.

Topic: Chiller tank level low

The chiller's display shows "Tank Level Low".

The chiller's coolant level has to be checked regularly. In case of too little amount of coolant the chiller may not work and cool properly anymore. Please add coolant to the chiller until the warning "Tank Level Low" disappears.

Spectra-Physics highly recommends using a mixture of **two parts of distilled water** and **one part of Clariant's Antifrogen L**. This mixture prevents the chiller and the laser's cooling channels from algae growth and corrosion. If Antifrogen L is not available in your country, it can be ordered from Spectra-Physics.

Topic: Remote connection

For remote control or check of the laser system a connection to the computer – on which the Spirit GUI is installed – needs to be established. Therefore, the computer has to be connected to the internet (via Wi-Fi

or LAN). Spectra-Physics uses the program **TeamViewer** for the remote connection that is pre- installed on all laptops of Spirit's delivery content. If another computer is used, please install version 8 of TeamViewer, which can be downloaded from <https://www.teamviewer.com/en/download/previous-versions/>.

Please contact service@spectra-physics.at to organize a remote session and report problems, warnings or error messages that have appeared. Our support team will inform you about the necessary safety instructions for a remote control of the laser system (e.g. laser safety, beam blockers, safety shutter, ...).

Topic: **Warranty extension**

Warranty extensions are available for all Spirit laser models. The standard warranty time is 12 months starting from delivery of the laser system. The standard warranty time can be **extended by 1, 2 or 3 years** and the **maximum total warranty time is 4 years**. The warranty extension needs to be ordered within the first 12 months after initial shipment (within the standard warranty time).

For a customized period of warranty or a warranty extension after the first 12 months after initial shipment, please contact sales@spectra-physics.at for a specific quote.

Topic: **Software "HMT" does not respond**

Please check, if the communication cable between laptop and controller is still connected properly. If the hardware connection is ensured, please click the "connect" button in the upper left corner of the GUI to re-connect. If the software seems frozen, please shut the Spirit® GUI down by the task manager and restart the GUI.

If the problem is not solved or appears too often, please contact service@spectra-physics.at and ask for a remote session to evaluate the root cause.

Topic: **No laser Output Power**

If the laser does not emit laser light, please check the following questions:

- Is the mechanical safety shutter opened?
- Is the interlock closed/short-cut (customer control cable)?
- Is the laser state "On" running (not "Off", "Standby" or "Initializing")?
- Is a "Power Setpoint [mW]" >0 mW selected?
- Is the "Shutter Open" check-box activated (= shutter open)?
- Is the "Pulse Enable" dropdown menu set to "internal" (= "always on")?
- Is the "Analog In" dropdown menu set to "disabled"?
- Are no error messages displayed on the GUI?

If all questions above can be answered with yes and the laser is still not emitting light, please contact service@spectra-physics.at.

Topic: **Internal powermeter shows 0 W output power**

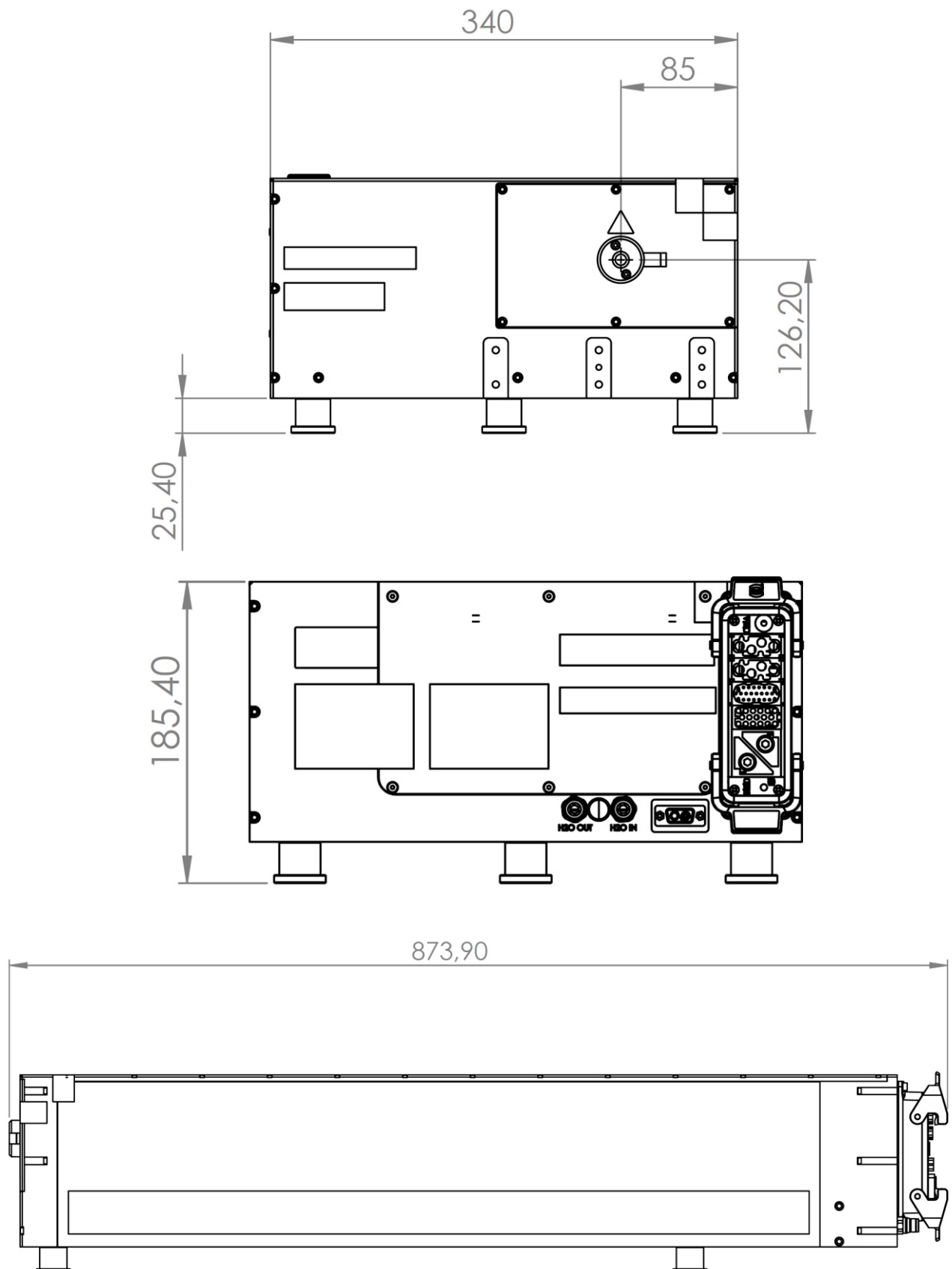
The internal powermeter read-out is 0 mW, although the laser state is "ON" and no error message is shown. This may be caused by the the following laser settings. Please check the following questions:

- Is a "Power Setpoint [mW]" >0 mW selected?
- Is the "Shutter Open" check-box activated (= shutter open)?
- Is the "Pulse Enable" dropdown menu set to "internal" (= "always on")?
- Is the "Analog In" dropdown menu set to "disabled"?

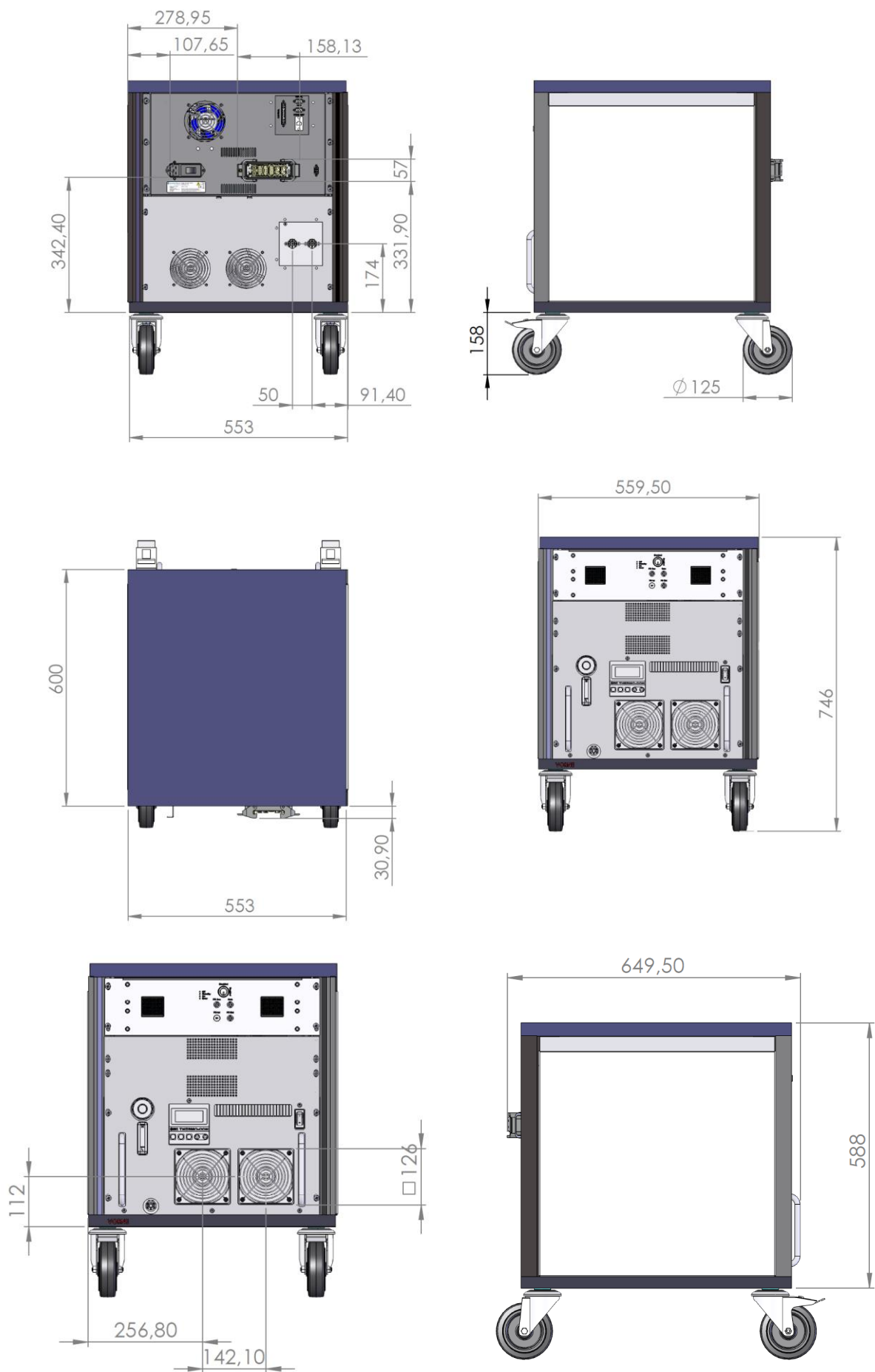
If all questions above can be answered with yes and the output power is still close to zero, please contact service@spectra-physics.at.

10 Specifications

10.1 System Drawings Laser Head



10.2 System Drawings Electronic Rack



10.3 Optical, Mechanical and Electrical Specifications 16 W HE-SHG

Parameter	Spirit 1040-16 HE	Spirit 1040-16 HE-SHG
Wavelength	(1040 ± 5) nm	(520 ± 3) nm
Bandwidth (FWHM)	<8 nm	<3 nm
Output Power ⁽¹⁾	>12 W (100 kHz), >16 W (≥200 kHz)	>6 W (100 kHz), >8 W (200 kHz), >4 W (500, 750, 1000 kHz)
max. Pulse Energy ⁽²⁾	120 μJ @ 100 kHz	<50% IR @ 100 kHz, 200 kHz
Repetition Rate	single pulse – 1 MHz	single pulse – 1 MHz
Pulse Width	<400 fs, nominal	<400 fs, nominal
Power stability	<1 % RMS (over 100 hours)	<1 % RMS (over 100 hours)
Spatial Mode	TEM ₀₀ with M ² < 1.2	TEM ₀₀ with M ² < 1.2
Beam Diameter	2.5 ± 0.5 mm (at laser exit)	2.5 ± 0.5 mm (at laser exit)
Divergence (full angle)	<1 mrad	<1 mrad
Pre Pulse Contrast Ratio	>250 : 1	>250 : 1
Polarization	horizontal	horizontal
Cold start time	<30 min	<30 min
Warm start time	<15 min	<15 min
Environmental specifications (during operation)		
Operating temperature	18 – 30 °C (64 – 86 °F)	18 – 30 °C (64 – 86 °F)
Relative humidity	<65 %, non-condensing	<65 %, non-condensing
Altitude (maximum)	3,000 m	3,000 m
Cooling requirements		
Laser Head	Closed loop chiller, included	Closed loop chiller, included
Power supply	Air cooled	Air cooled
Utility Requirements		
Voltage	100 – 240 V~, 50/60 Hz	100 – 240 V~, 50/60 Hz
Current	<14 A	<14 A
Laser Head Physical Characteristics		
Dimensions (L, W, H)	32.5 x 14 x 6.5" (875 x 357 x 165 mm ³)	32.5 x 14 x 6.5" (875 x 357 x 165 mm ³)
Weight	155 lbs (70 kg)	155 lbs (70 kg)
Power supply Physical Characteristics		
Dimensions (L, W, H)	23.6 x 21.8 x 29.3" (600 x 553 x 746 mm ³)	23.6 x 21.8 x 29.3" (600 x 553 x 746 mm ³)
Weight	176 lbs (80 kg)	176 lbs (80 kg)

(1) Maximum conversion efficiency of 50% for SHG is achieved at 100 kHz and 200 kHz; lower efficiency for remaining basic repetition rates will occur, but always >25 %.

(2) Highest pulse energy is available at 100 kHz; the output power for fundamental wavelength is >12 W for 100 kHz and >16 W for all higher (optional) basic repetition rates (200, 500, 750 and 1000 kHz).

10.4 Optical, Mechanical and Electrical Specifications 30 W HE-SHG

Parameter	Spirit 1040-30 HE	Spirit 1040-30 HE-SHG
Wavelength	(1045 ± 5) nm	(523 ± 3) nm
Bandwidth (FWHM)	<8 nm	<3 nm
Output Power ⁽³⁾	>30 W	>15W (250kHz), >10W (500, 750, 1000 kHz)
max. Pulse Energy ⁽⁴⁾	> 120 μJ @ 250kHz	<50% IR @ 250kHz
Repetition Rate	single pulse – 1 MHz	single pulse – 1 MHz
Pulse Width	<400 fs, nominal	<400 fs, nominal
Power stability	<1 % RMS (over 100 hours)	<1 % RMS (over 100 hours)
Spatial Mode	TEM ₀₀ with M ² < 1.2	TEM ₀₀ with M ² < 1.2
Beam Diameter	2.5 ± 0.5 mm (at laser exit)	2.5 ± 0.5 mm (at laser exit)
Divergence (full angle)	<1 mrad	<1 mrad
Pre Pulse Contrast Ratio	>250 : 1	>250 : 1
Polarization	horizontal	horizontal
Cold start time	<30 min	<30 min
Warm start time	<15 min	<15 min
Environmental specifications (during operation)		
Operating temperature	18 – 30 °C (64 – 86 °F)	18 – 30 °C (64 – 86 °F)
Relative humidity	<65 %, non-condensing	<65 %, non-condensing
Altitude (maximum)	3,000 m	3,000 m
Cooling requirements		
Laser Head	Closed loop chiller, included	Closed loop chiller, included
Power supply	Air cooled	Air cooled
Utility Requirements		
Voltage	100 – 240 V~, 50/60 Hz	100 – 240 V~, 50/60 Hz
Current	<14 A	<14 A
Laser Head Physical Characteristics		
Dimensions (L, W, H)	32.5 x 14 x 6.5" (875 x 357 x 165 mm ³)	32.5 x 14 x 6.5" (875 x 357 x 165 mm ³)
Weight	155 lbs (70 kg)	155 lbs (70 kg)
Power supply Physical Characteristics		
Dimensions (L, W, H)	23.6 x 21.8 x 29.3" (600 x 553 x 746 mm ³)	23.6 x 21.8 x 29.3" (600 x 553 x 746 mm ³)
Weight	176 lbs (80 kg)	176 lbs (80 kg)

⁽³⁾ Maximum conversion efficiency of 50% for SHG is achieved at 250kHz; lower efficiency for remaining basic repetition rates will occur, but always >25 %.

⁽⁴⁾ Highest pulse energy is available at 250kHz; the output power for fundamental wavelength is >30 W for 250kHz, 500, 750 and 1000 kHz.

11 Appendix A1: Spirit CANopen Communication Overview

System / Subsystem	Command	Description	Units		Setpoints		Default values	Object Dictionary		Access		Data type
			Unit	Dec. places	min	max		Index hex	Subindex hex	PDO/SDO	RW/RO	
System	Common	Control		0	0	2		2000	0	PDO	RW	UINT8
		Status		0				2001	0	PDO	RO	UINT8
		Warnings		0				2100	0	PDO	RO	UINT16
		Errors		0				2101	0	PDO	RO	UINT16
		Master Off Shutdown Time	s	3	0	60		2005	0	SDO	RW	UINT16
		Baudrate (Restart the device after changing the baudrate)		0	0	7		2004	0	SDO	RW	BYTE
		Runtime	min	0				2002	0	SDO	RO	UINT32
		OnOff Statistic		0				2003	0	SDO	RO	UINT32
		Article Number		0				2006	0	SDO	RO	UINT32

System / Subsystem			Command	Description	Units		Setpoints		Default values	Object Dictionary		Access		Data type
					Unit	Dec. places	min	max		Index hex	Subindex hex	PDO/ SDO	RW/ RO	
System	Seeder		Warnings	bit 0 (dec 1): Diode Cur bit 1 (dec 2): Diode Cur Volt bit 2 (dec 4): Diode Temp bit 3 (dec 8): Diode Temp Cur bit 4 (dec 16): Diode Temp Cur Volt bit 5 (dec 32): Xtal Temp bit 6 (dec 64): Xtal Temp Cur bit 7 (dec 128): Xtal Temp Cur Volt bit 8 (dec 256): ML Signal bit 9 (dec 512): ML Detector		0				2104	0	SDO	RO	UINT16
			Errors	see above		0				2105	0	SDO	RO	UINT16
			Frequency Act	Measured oscillator frequency	kHz	1				231C	0	SDO	RO	INT16
		Diode	Cur Act	Active laser diode current	A	3				2301	0	SDO	RO	UINT16
			Temp Act	Active temperature of the laser diode	°C	2				2303	0	SDO	RO	INT16
		Xtal	Temp Act	Active temperature of the crystal	°C	2				2306	0	SDO	RO	INT16
		Modelock Detector	Seeder Power Act	Active seeder signal level	dBm	1				2311	0	SDO	RO	INT16
	Amplifier		Warnings	bit 0 (dec 1): Diode Cur bit 1 (dec 2): Diode Cur Volt bit 2 (dec 4): Diode Temp bit 3 (dec 8): Diode Temp Cur bit 4 (dec 16): Diode Temp Cur Volt bit 5 (dec 32): Xtal Temp bit 6 (dec 64): Xtal Temp Cur bit 7 (dec 128): Xtal Temp Cur Volt		0				2106	0	SDO	RO	UINT16
			Errors	see above		0				2107	0	SDO	RO	UINT16
		Diode	Cur Act	Active laser diode current	A	3				2309	0	SDO	RO	UINT16
			Temp Act	Active temperature of the laser diode	°C	2				230B	0	SDO	RO	INT16
		Xtal	Temp Act	Active temperature of the crystal	°C	2				230E	0	SDO	RO	INT16

System / Subsystem			Command	Description	Units		Setpoints		Default values	Object Dictionary		Access		Data type
					Unit	Dec. places	min	max		Index hex	Subindex hex	PDO/ SDO	RW/ RO	
System	Switching Unit		Warnings	bit 0 (dec 1): PDG power error bit 1 (dec 2): PDG I-Lock error bit 2 (dec 4): PDG seed level error bit 3 (dec 8): PDG frequency error bit 5 (dec 32): PDG gate error		0				2108	0	SDO	RO	UINT16
			Errors	not used		0				2109	0	SDO	RO	UINT16
		Pulse Picker Divider	Pulse Picker Divider	Integer Divider		0	0	10000		2501		SDO	RW	UINT32
		Analog In Enable	Switch between Power control via Analog In and HMI (Fast Attenuation Mode)	0: Disable 1: Enable		0	0	1		2505	0	SDO	RW	BYTE
		Pulse Enable Set	Set Pulse Enable	0: Pulse Picker External enable 1: Pulse Picker Internal enable		0				2506	0	SDO	RW	UINT16
		Pulse Enable	Act	0: Pulse disable 1: Pulse enabled	V	0				231E	0	SDO	RO	UINT16
		Analog In Level	Act	Voltage at Analog In Port		2	0	5		2321	0	SDO	RO	UINT16
		Reprate	Reprate Switch	Switching between Reprates 0: Reprate 0 1: Reprate 1 2: Reprate 2 3: Reprate 3 4: Reprate 4		0	0	4		2502	0	SDO	RW	BYTE
	Power Control		Warnings	bit 0 (dec 1): Shutter bit 1 (dec 2): Attenuator callibrated bit 2 (dec 4): Signal LED		0				210A	0	SDO	RO	UINT16

				Voltage									
			Errors	see above		0			210B	0	SDO	RO	UINT16
		Attenuator	SP	Attenuator setpoint, 0%...min. attenuation 100%...max. attenuation	%	0	0	100	2500	0	SDO	RW	UINT8
		Power	SP	Power setpoint in W /Optional	W	3	0	60	2509	1	SDO	RW	UINT16
		Custom Shutter	SP	0: Close shutter 1: Open shutter		0	0	1	2504	0	SDO	RW	BYTE
		PowerCtrl Modus	Harmonic Conversation (Wavelength Mode)	0: Infrared(1040nm 1: Green /Optional/520nm 5: UV /Optional		0	0	5	2508		SDO	RW	BYTE
		Shutter	Act	0: Shutter closed 1: Shutter open		0			2313	0	SDO	RO	BYTE
		Signal LED	Act	Voltage of Signal LED	V	3			2503		SDO	RO	INT16
		Powermeter	Act	Active power at output	W	2			2312	0	SDO	RO	INT16

System / Subsystem		Command	Description	Units		Setpoints		Default values	Object Dictionary		Access		Data type
				Unit	Dec. places	min	max		Index hex	Subindex hex	PDO/ SDO	RW/ RO	
System	Electronics	Warnings	bit 0 (dec 1): Supply voltage bit 1 (dec 2): Electronics Temp bit 2 (dec 4): DSAMP Head Temp bit 3 (dec 8): DSOSC Head Temp bit 4 (dec 16): PSXTAL Head Temp bit 8 (dec 256): CHILLER COM bit 9 (dec 512): DSOSC COM bit 10 (dec 1024): DSAMP COM bit 11 (dec 2048): PSXTAL COM bit 12 (dec 4096): PDG COM bit 13 (dec 8192): HV bit 14 (dec 16384): Baudrate switch bit 15 (dec 32768) ATTEN COM		0				210C	0	SDO	RO	UINT16
		Errors	see above		0				210D	0	SDO	RO	UINT16
		Supply Act	Supply Voltage	V	2				2314	0	SDO	RO	INT16
		Temp Act	Electronics Temperature	°C	2				2315	0	SDO	RO	INT16
		DSAMP Status Act	DSAMP Status		0				2316	0	SDO	RO	UINT16
		DSAMP Temp Act	DSAMP Head Temp	°C	2				2317	0	SDO	RO	INT16
		DSOSC Status Act	DSOSC Status		0				2318	0	SDO	RO	UINT16
		DSOSC Temp Act	DSOSC Head Temp	°C	2				2319	0	SDO	RO	INT16
		PSXTAL Status Act	PSXTAL Status		0				231A	0	SDO	RO	UINT16
		PSXTAL Temp Act	PSXTAL Head Temp	°C	2				231B	0	SDO	RO	INT16
		Ctrl AMP Diode Error	AMP Modul State		0				2320	0	SDO	RO	UINT16

12 Appendix A2: Spirit TCP/IP Communication Overview

Submodule	Parameter	Description	Unit	Decimals	RW ³ /RO ⁴	Type	Min	Max
Common	<i>state_control</i>	0: Laser Off 1: Laser Standby 2: Laser On		0	RW	UINT8	0	2
	<i>state_actual</i>	0: Initializing, 1: No Operation 2: Warming Up, 3: Init Laser Off 4: Laser Off, 5: Init Laser Standby 6: Laser Standby, 7: Init Laser On 8: Laser On, 9: Manual Mode 10: Init Error, 11: Error		0	RO	UINT8		
	<i>common_warnings</i>	bit 0 (dec 1): Interlock is opened bit 1 (dec 2): Chiller bit 2 (dec 4): Seeder bit 3 (dec 8): Amplifier bit 4 (dec 16): Switching Unit bit 5 (dec 32): Power Control bit 6 (dec 64): Controller / Electronics bit 7 (dec 128): System Error bit 8 (dec 256): Second Harmonic Generation (SHG) bit 9 (dec 512): fs2ps Stage bit 13 (dec 8192): MasterOff external bit 14 (dec 16384): Keyswitch (only warning) bit 15 (dec 32768): MasterOff internal		0	RO	UINT16		
	<i>common_errors</i>	see above		0	RO	UINT16		
	<i>master_off_shutdown_time</i>	0: inactive >0: if no SDOs or PDOs are received in this interval the laser will be turned off	s	3	RW	UINT16	0	65.535

³ Read/Write

⁴ Read Only

Submodule	Parameter	Description	Unit	Decimals	RW ³ /RO ⁴	Type	Min	Max
Chiller	<i>temperature_control_warnings</i>	bit 0 (dec 1): Temperature bit 1 (dec 2): Fluid Level limit reached		0	RO	UINT16		
	<i>temperature_control_errors</i>	bit 0 (dec 1): Temperature bit 1 (dec 2): Fluid Level limit reached bit 2 (dec 4): Chiller flow		0	RO	UINT16		
Seeder	<i>seeder_warnings</i>	bit 0 (dec 1): Diode Current bit 1 (dec 2): Diode Voltage bit 2 (dec 4): Diode Temperature bit 8 (dec 256): Absolute Modelock limit bit 9 (dec 512): Relative Modelock limit		0	RO	UINT16		
	<i>seeder_errors</i>	see above		0	RO	UINT16		
Amplifier	<i>amplifier_warnings</i>	bit 0 (dec 1): Diode Current bit 1 (dec 2): Diode Voltage bit 2 (dec 4): Diode Temperature bit 5 (dec 32): Crystal Temperature		0	RO	UINT16		
	<i>amplifier_errors</i>	see above		0	RO	UINT16		
Switching Unit	<i>switching_unit_warnings</i>	bit 0 (dec 1): PDG Supply Voltage bit 1 (dec 2): PDG Interlock bit 2 (dec 4): PDG Seed Level bit 3 (dec 8): PDG Frequency bit 5 (dec 32): PDG Gate Error		0	RO	UINT16		
	<i>switching_unit_errors</i>	see above		0	RO	UINT16		
	<i>pulsepicker_divider</i>	Divider		0	RW	UINT32	0	1e6
	<i>analog_in_enable</i>	0: Disabled 1: Enabled		0	RW	UINT8	0	1
	<i>pulse_enable_mode</i>	0: External 1: Always On 2: Differential		0	RW	UINT8	0	2
	<i>pulse_enable_actual</i>	0: Pulse disable 1: Pulse enabled		0	RO	UINT8		

Submodule	Parameter	Description	Unit	Decimals	RW ³ /RO ⁴	Type	Min	Max
	<i>repetition_rate</i>	Switching between Repetition Rates 0: Repetition Rate 0 1: Repetition Rate 1 2: Repetition Rate 2 3: Repetition Rate 3 4: Repetition Rate 4		0	RW	UINT8	0	4
	<i>repetition_rate_actual</i>	Repetition Rate	kHz	1	RO	Float		
Power Control	power_control_warnings	bit 0 (dec 1): Shutter bit 3 (dec 8): Safety Shutter Relay open bit 4 (dec 16): Power Maximum reached		0	RO	UINT16		
	power_control_errors	see above		0	RO	UINT16		
	power_setpoint_percentage	Attenuator setpoint (Transmission ratio) 0%...min. attenuation 100%...max. attenuation	%	0	RW	UINT8	0	100
	power_setpoint	Power setpoint	W	3	RW	Float		
	shutter_control	0: Close shutter 1: Open shutter		0	RW	UINT8	0	1
	shutter_state	0: Shutter closed 1: Shutter open		0	RO	UINT8		
	power_actual	Actual output power	W	3	RO	Float		
	shutter_relay_state	0: Safety Shutter Relay open 1: Safety Shutter Relay closed		0	RO	UINT8		

Submodule	Parameter	Description	Unit	Decimals	RW ³ /RO ⁴	Type	Min	Max
Controller	controller_warnings	bit 0 (dec 1): Supply Voltage (24V) bit 1 (dec 2): Electronics Temperature LCR1 bit 2 (dec 4): Amplifier Laser Diode TEC Dev. Temp. bit 3 (dec 8): DSOSC Device Temperature bit 4 (dec 16): PSXTAL Device Temperature bit 5 (dec 32): HV Communication bit 6 (dec 64): Amplifier Laser Diode Driver Comm. bit 8 (dec 256): Chiller Communication bit 9 (dec 512): DSOSC Communication bit 10 (dec 1024): Amplifier Laser Diode TEC Comm. bit 11 (dec 2048): PSXTAL Communication bit 12 (dec 4096): PDG Communication bit 13 (dec 8192): HV Error bit 15 (dec 32768): SHG Attenuator Comm. bit 16 (dec: 65536): DSOSC Internal Error bit 17 (dec: 131072): PSXTAL Internal Error bit 18 (dec: 262144): Amplifier LD TEC Internal Error bit 19 (dec: 524288): Amplifier LD Driver Internal Error bit 20 (dec 1048576): fs2ps Stage Communication		0	RO	UINT32		
	controller_errors	see above		0	RO	UINT32		

Submodule	Parameter	Description	Unit	Decimals	RW ³ /RO ⁴	Type	Min	Max
Controller	system_error	0: No System Error present		0	RO	UINT8		
		2: Timeout initializing the system						
		3: Timeout reaching the initialized temperatures						
		4: Timeout initializing the Laser-Off state						
		5: Timeout initializing the Laser-Standby state						
		6: Timeout initializing the Laser-On state						
		7: Too many MQ-protocol-timeouts on the RS232-bus						
		8: LED error						
		10: More than 15 EEPROM write-cycles per minute						
		11: Configurator error						
		12: Data corruption of both copies (defaults loaded)						
		13: Data corruption of first copy (loaded from valid second copy)						
		14: Data corruption of second copy (loaded from valid first copy)						
		15: Update process error (defaults loaded)						
		16: Unsupported configurator						
		17: Data size differs (defaults loaded)						
		18: Configurator update successful						
		19: Memory access violation (read or write outside accessible memory area)						
		20: Corrupted data size						
		21: Too many HQ-protocol-timeouts on the default baudrate of 4800						
		22: Too many HQ-protocol-timeouts on the fast baudrate of 125000						
		23: Too many MQ-protocol-timeouts on the RS422-bus						
		24: Power-on reset						
		25: External reset						
		26: Brown-out reset						
		27: Watchdog reset						
		28: JTAG reset						
		29: Too many Modbus-protocol-timeouts (used for the SMC-chiller)						
		30: HVQ-Error						
		31: HV-device temperature out of range						
		32: HV-supply voltage error or input frequency error						
		33: Gate-signal did not go to 0 on time						

Submodule	Parameter	Description	Unit	Decimals	RW ³ /RO ⁴	Type	Min	Max
SHG	shg_warnings	bit 0 (dec 1): SHG Crystal Temperature bit 5 (dec 32): Air Purging Unit (Filter)		0	RO	UINT16		
	shg_errors	see above		0	RO	UINT16		
	wavelength_setpoint	Wavelength (0=IR, 1=Green)			RW	UINT8		

13 APPENDIX A3: User Control Interface

The User Control Interface is a 21WA4 mixed Sub-D female connector with three BNC connectors for additional functionality. The Interface is used to provide several Input and Outputs to the customer.

Pin	Connection	User Control Cable drawing
A1	SYNC Out (BNC)⁵	The diagram illustrates the wiring for the User Control Interface. It features a Sub-D 21WA4 (M) connector with pins 1 through 17. Three BNC connectors are shown: X2.1 (Pulse Enable), X2.2 (Analog In), and X2.3 (SYNC). The connections are as follows: Pin 1 connects to X2.1, Pin 2 to X2.2, Pin 3 to X2.3, and Pin 4 is shorted to ground. Pins 5 through 17 are not connected to any external components in this diagram.
A2	Pulse Enable (BNC)	
A3	Analog In (BNC)	
A4	<i>not connected</i>	
1	I-Lock +	
2	Custom Error+⁶	
3	Shutter Status K1.1⁷	
4	Shutter Status K1.2⁷	
5	Pulse Enable Diff. RS485 Rx +	
6	Shutter Status K2.1⁷	
7	Shutter Status K2.2⁷	
8	+24V	
9	Shutter Relay 24V	
10	I-Lock –	
11	Custom Error-⁶	
12	Pulse Enable Diff. RS485 Rx –	
13	<i>not connected</i>	
14	GND	
15	GND	
16	GND	
17	Shutter Relay	

⁵ TTL Output Signal.

⁶ Isolated optocoupler output. Maximum collector current: 10mA. Maximum collector emitter voltage: 40V. Closed when no error is present and timing generation is enabled.

⁷ The shutter connection pins relate to the shutter relay model (type: PSR-SCP- 24DC/ESD/4X1/30 Phoenix Contact). For more detailed information please refer to the related user manual available on the supplier webpage.

14 APPENDIX A4: China RoHS

The following products do currently not fulfil the China RoHS directive SJ/T 11363-2006:

部件名称	有害物质					
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr(VI))	多溴联苯 (PBB)	多溴联苯醚 (PBDE)
06883	X					
07758	X					
10216	X					
11931	X					
08163	X					
06344	X					

此表是按照 SJ/T 11364 的规定编制。

O : 表示该部件的所有均质材料中所含该有害物质均低于 GB/T 26572 的规定。

X : 表示该部件所使用的至少一种均质材料中所含该有害物质高于 GB/T 26572 的规定。

(企业可根据其实际情况在空格中对“X”标记进一步提供技术说明。)



15 Notes
